

IS THERE A PLATONIC TREE?

CALIBRATING LATENT HYPERBOLICITY

IN FOUNDATION MODELS

Anonymous authors

Paper under double-blind review

ABSTRACT

Hyperbolic representation learning rests on a premise we call latent hyperbolicity: standard models are already tree-like because their class geometry scores a low Gromov δ , a measure of tree-likeness that is zero for a tree. That low value is also what a random cloud of the same dimension and spectrum scores, so we build an instrument that reads every score as its excess over 200 such clouds, and add a depth test whose power is measured. We apply it to 12 vision backbones, 6 datasets and 15 text models. On image features, where the premise is read, the calibrated reading is indistinguishable from a random cloud in most cells and small elsewhere. Class centroids do carry structure that a random cloud of the same shape does not, in 44 of 72 model–dataset cells, but so does a star of clusters. The depth test certifies in 4 of 12 ImageNet backbones that clusters are oriented toward their hubs, not that the hubs form a hierarchy: randomizing the orientations removes every verdict, whereas an implanted hierarchy, synthetic or trained in, survives it. No hub hierarchy is found in the nine backbones where the decoupled control detects an implanted one, and the test is blind in the other three. A backbone trained in hyperbolic space, MERU, shows the same structure as its Euclidean twin and lives in the near-flat regime. Across models the trees are moderately shared: the naive comparison manufactures an island, every recipe partially recovers the human taxonomy once the clustering cut is controlled, and the self-supervised tree is angular. Read correctly, foundation models organize classes into clustered structure that is moderately shared and does not converge to one common tree; no hub hierarchy is found where the test has power; and their raw tree-likeness is not evidence for hyperbolic geometry.

1 INTRODUCTION

Semantic knowledge is hierarchical, and hyperbolic space is the natural geometric home for hierarchies (Nickel & Kiela, 2017; Sarkar, 2011; Ganea et al., 2018; Chami et al., 2019). A line of work therefore imposes it on representations at training time, from vision to vision–language and language (Khrulkov et al., 2020; Ermolov et al., 2022; Atigh et al., 2022; Bdeir et al., 2024; Desai et al., 2023; Pal et al., 2025; Chen et al., 2022; He et al., 2025; Sinha et al., 2024). These methods cite a premise, which we call *latent hyperbolicity*: standard models already organize concepts in a tree-like metric structure. The evidence is a single number, the Gromov δ of their features, which is zero for a metric tree and grows as a metric departs from one. Measured on CNN and ViT features (Khrulkov et al., 2020; Bdeir et al., 2024), on token embeddings (Yang et al., 2025) and on word vectors (Tifrea et al., 2019), it comes out low. Those readings are taken at the sample level, on individual images or tokens rather than on classes, and they are never calibrated: a value is called low on its own, without asking what a structureless cloud of the same dimension and spectrum would score. Is the premise right, and what does it license? Figure 1 frames the question: an observer who sees only a shadow, the raw δ , asks which of three worlds cast it, a structureless cloud, a star of clusters or a tree. All three read alike.

Why is the reading low? Gromov δ assigns zero to trees (Gromov, 1987), but its reference level depends on dimension, spectrum and statistic, so a raw value cannot be called low on its own. High-dimensional clouds concentrate toward equidistance (Beyer et al., 1999; Aggarwal et al., 2001), which is a degenerate star tree: we call this the *dimension confound*, the geometric counterpart of the width confounder of Gröger et al. (2026). Anisotropic spectra lower the effective dimension and raise the reading: the *spectrum confound*. And the supremum over sampled quadruples grows

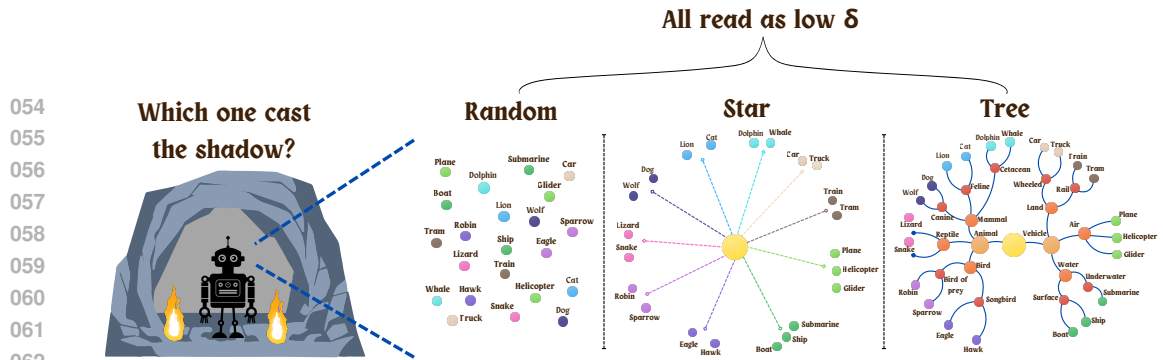


Figure 1: **Plato’s cave: a random cloud, a star and a tree cast the same shadow.** An observer who sees only shadows cannot tell which object cast them, and a raw δ is such a shadow: points with no structure, clusters around a hub, and a nested hierarchy all read as low δ . The instrument reads the objects instead of the shadow: no structure beyond the null in the cloud, clusters without depth in the star, clusters with depth in the tree.

with the budget and does not converge (Fournier et al., 2015): the *statistic confound*. The three are visible in real models: GPT-2 M sits at its matched null despite a low raw reading, and classic sentence embedders share their raw reading with vision-contrastive models while carrying none of their structure.

We build the instrument that reads δ correctly. It calibrates the estimator on reference geometries, expresses every reading as an *excess* over a null cloud matched in dimension and covariance spectrum, ranks that excess against matched replicates, and adds a *depth test* whose false-alarm rate and power are measured on real clouds. With it we ask of geometry what convergence work asks of similarity (Huh et al., 2024; Gröger et al., 2026): what survives calibration, and is it shared? They calibrate similarity across models; we calibrate geometry within a model, and comparing trees across models needs a calibration of its own.

The answer has three parts. On image features, where the premise is read, the calibrated reading is indistinguishable from a random cloud of the same shape in most cells and small elsewhere; the four values reported by Khurlov et al. (2020) are reproduced and, calibrated, two of them are indistinguishable from a random cloud. On class centroids there is structure that a random cloud does not have, in 44 of 72 cells and 30 of 36 on datasets with 47 classes or more, but a star of clusters already produces it. The depth test certifies in 4 of 12 ImageNet backbones that clusters are oriented toward their hubs, not that the hubs form a hierarchy: randomizing the orientations removes every verdict, whereas an implanted hierarchy, synthetic or trained in, survives it; no hub hierarchy is found in the nine backbones where the decoupled control detects an implanted one, and the test is blind in the other three. A backbone trained in hyperbolic space shows the same structure as its Euclidean twin and lives in the near-flat regime. Across models the trees are moderately shared: the naive comparison manufactures an island, every recipe partially recovers the human taxonomy once the clustering cut is controlled, and the self-supervised tree is angular.

Our contributions are:

1. **The instrument.** Estimator calibration, spectrum-matched nulls read as corrected ranks, and a depth test with measured false-alarm rate and power.
2. **The census.** Structure beyond a random cloud in most models, clusters oriented toward their hubs in a few, no hub hierarchy in the nine backbones where the decoupled control detects an implanted one, a blind test in the other three, and a nominally hyperbolic backbone as the control for imposing the geometry.
3. **The map of trees.** The island is an artifact of the cut; sharing is graded and supervision-dependent, and the self-supervised tree is angular.
4. **Consequences for practice.** What to measure before imposing curvature, and which zero-cost readout collects what is there.

2 RELATED WORK

Reading δ against a matched reference has precedent in network science, where the δ of real graphs is compared with that of random graphs of comparable size (Narayan & Sanjeev, 2011; Kennedy et al., 2013; Adcock et al., 2013); we bring the idea to embedding clouds, where the reference must match dimension and spectrum. The curvature itself has been studied as a representation tradeoff (Sala et al., 2018) and as a mixed-curvature product to be learned (Gu et al., 2019), which presupposes a diagnostic of the kind we calibrate. The analyses of learned representations characterize Euclidean structure and do not test for tree-likeness: representation similarity, intrinsic dimension

and neural collapse (Kornblith et al., 2019; Ansuini et al., 2019; Pope et al., 2021; Pappas et al., 2020). Closest to us is the work on representational convergence: the Platonic Representation Hypothesis (Huh et al., 2024) argues that models converge toward shared representations, and its cross-modal convergence has been stress-tested at scale (Koeperke et al., 2026); Gröger et al. (2026) show that permutation-calibrated similarity removes most of that convergence while local neighborhood structure survives, and Park et al. (2024) read concept hierarchy in language models from the orthogonality of concept directions. We are the geometric complement of this program: the same calibration applied within a model instead of across models, with a depth test measured on real clouds, a hyperbolic backbone as the control for imposing the geometry, and a calibration of the tree-to-tree comparison itself.

3 METHODOLOGY

3.1 BACKGROUND: GROMOV δ

Tree-likeness is measured on quadruples. Take any four points w, x, y, z of a metric space with distance d . They can be paired in three ways, and each pairing has a sum of two distances:

$$d(w, x) + d(y, z), \quad d(w, y) + d(x, z), \quad d(w, z) + d(x, y). \quad (1)$$

Sort the three sums as $S_{\max} \geq S_{\text{mid}} \geq S_{\min}$. In a tree the two largest sums are always equal, because the two pairings that cross the trunk traverse the same path, so the four-point defect $(S_{\max} - S_{\text{mid}})/2$ is zero; the further a metric is from a tree, the larger the defect can become. Gromov δ is the worst case, the supremum of the defect over all quadruples (Gromov, 1987). A metric tree has $\delta = 0$. Euclidean space has no bound at all: enlarging a square enlarges its defect in proportion. Hyperbolic space sits in between, with a bounded defect whose value for the hyperbolic plane is known, $\ln 2$ under this definition (Nica & Špakula, 2016).

Definition 1 (four-point defect). For a quadruple (w, x, y, z) with the pairing sums of Eq. 1 sorted as $S_{\max} \geq S_{\text{mid}} \geq S_{\min}$, the four-point defect is $\delta(w, x, y, z) = \frac{1}{2} (S_{\max} - S_{\text{mid}})$, and the Gromov δ of a metric space is the supremum of the defect over its quadruples.

3.2 ESTIMATION AND NORMALIZATION

Three steps turn δ into a number that can be compared across models. First, a thousand centroids have far more quadruples than can be enumerated, so the defect is computed on half a million random quadruples. Second, the largest defect in that sample is a poor estimate of the supremum: it belongs to a single extreme quadruple and keeps growing as more quadruples are drawn (Fournier et al., 2015). We therefore read the 99.9th percentile of the sampled defects, $\hat{\delta}_{99.9}$, the value that 99.9 per cent of them fall below: it keeps the tail of the defects, as the supremum does, but it is set by hundreds of quadruples rather than by one. It also does not move with the sample size: across quadruple budgets the ImageNet reading drifts by less than half its own spread (Table 2). Third, the defect grows with the size of the point set, so we divide by the diameter, the largest pairwise distance:

$$\delta_{\text{norm}} = \frac{\hat{\delta}_{99.9}}{\text{diam}}. \quad (2)$$

On this scale a tree reads zero, and a hyperbolic region of radius four or more reads below a sphere (Table 3).

Definition 2 (reading of record). The reading of record of a cloud is δ_{norm} of Eq. 2, with $\hat{\delta}_{99.9}$ computed over half a million sampled quadruples per seed and averaged over ten seeds.

Proposition 1 (range bound and dimension confound). Let $d_{\min}$ and $d_{\max}$ be the smallest and the largest pairwise distance of a finite metric space. (a) For any quadruple the four-point defect is at most $d_{\max} - d_{\min}$, hence $\delta_{\text{norm}} \leq 1 - d_{\min}/d_{\max}$. (b) For an iid Gaussian cloud of n points in dimension d , with n fixed, $d_{\min}/d_{\max} \rightarrow 1$ as d grows (Beyer et al., 1999; Aggarwal et al., 2001), so $\delta_{\text{norm}} \rightarrow 0$ without any structure.

A raw value cannot be called low on its own. The reference level of the reading depends on dimension, spectrum and statistic. High dimension lowers it (the dimension confound), as Proposition 1, proved in Appendix A, states for a cloud with no structure at all. Anisotropy raises it through the effective dimension (the spectrum confound), and a sampled supremum grows with the budget (the statistic confound). The calibration on reference geometries confirms this: the estimator recovers the tree zero and the $\ln 2$ of the hyperbolic plane, spheres stay high, and iid Gaussians drift down with the dimension (Table 3).



Figure 2: **The raw reading cannot be read on its own.** (a) Raw reading and matched null per ImageNet backbone, ordered by dimension, with the isotropic Gaussian reference dashed; the gap between the two bars is the excess. (b) Same dataset, same reading, opposite verdict: ViT-B on CIFAR-100 images and DINO-B on CIFAR-100 centroids.

3.3 SPECTRUM-MATCHED NULL, EXCESS AND GENUINENESS

Definition 3 (Haar null). Let $X = U\Sigma V^\top$ be the singular value decomposition of the centered cloud. A null replicate is $X' = Q\Sigma V^\top$ with Q the orthogonal factor of a column-centered Gaussian matrix, orthogonal to the all-ones vector and Haar-distributed on the centered subspace, so X' has exactly the centered singular values of X and structureless coefficients.

The null is a cloud with the real shape and no structure. Gaussian coefficients reproduce the spectrum only in expectation, fall short of it when n is small relative to d , and inflate the excess of low-rank clouds. The centered Haar construction reproduces the centered spectrum exactly, so it is the record. The spectrum null conditions on the second moments, to which a hierarchy contributes, so the excess is conservative and a low excess does not exclude hierarchy carried by the spectrum. The uncentered and the Gaussian readings are reported beside it (Table 4).

Definition 4 (excess). With K null replicates $X'_1, \dots, X'_K$ of Definition 3, the excess of a cell is

$$e(X) = \delta_{\text{norm}}(X) - \frac{1}{K} \sum_{k=1}^K \delta_{\text{norm}}(X'_k). \quad (3)$$

The rank of the reading is the evidence. The excess is a size. The evidence is the rank of the reading among its replicates and the left-tail p that the rank gives,

$$r = \#\{k : \delta_{\text{norm}}(X'_k) > \delta_{\text{norm}}(X)\}, \quad p = \frac{1 + K - r}{K + 1}, \quad K = 200. \quad (4)$$

Rank-based calibration gives the score its finite-sample validity.

Definition 5 (genuine). Order the m left-tail p -values of a census as $p_{(1)} \leq \dots \leq p_{(m)}$. A model-dataset cell is genuine when its p is at or below the Benjamini–Hochberg threshold at level five per cent,

$$p \leq p_{(i^*)}, \quad i^* = \max\{i : p_{(i)} \leq 0.05 i/m\}, \quad (5)$$

and the word is used in this sense only.

A negative genuine excess means structure beyond the second moments. A negative genuine excess certifies structure beyond the second moments; clustering is its most plausible reading, supported by the superclass recovery of Section 5.4. The word tree-like is reserved for the depth test of Section 3.4. The null moves with the dimension, so the reading is the gap between the raw bar and the null. Two cells on the same dataset with the same reading can receive opposite verdicts (Figure 2).

3.4 DEPTH TEST

Definition 6 (hub null and excess B). Given a frame that assigns each centroid x_i to a cluster $k(i)$ with hub $h_{k(i)}$, the cluster mean, write $x_i = h_{k(i)} + o_i$. The hub null keeps every offset o_i and replaces the hubs by a Haar resample with their spectrum, $x'_i = h'_{k(i)} + o_i$, and the excess of Eq. 3 against this null, e_B , measures structure above the clusters of the frame only.

Definition 7 (matched star). The matched star of a cloud keeps the real clusters, each as a Haar sample with the cluster’s own covariance, and places their hubs as Gaussian draws at the real hub

radius: the clustering of the real cloud with no hierarchy above it. Its spectrum-matched-hub variant draws the hubs as a Haar resample of the real hubs.

The depth test compares the real cloud with its matched star. With e_B read on the real cloud and on its matched star, the depth statistic and its standardized form are

$$\text{depth} = e_B(X) - e_B(X_{\text{star}}), \quad z = \text{depth} / \text{s.d.}, \quad (6)$$

The spread is taken over star seeds and null replicates, and a cloud is certified against its matched star when $z \leq -2$. The test is needed because a star with no hierarchy passes the census of Definition 5. The frame is the WordNet cut on ImageNet and the coarse labels on CIFAR-100, and the false-alarm rate and power are measured on real clouds in Section 5.3.

3.5 COMPARING TREES ACROSS MODELS

Trees are compared under a configuration chosen before the comparison. Each model’s hierarchy is a dendrogram over the shared class centroids. Two hierarchies are compared by the adjusted Rand index between their cut partitions, by the correlation of their cophenetic distances, and by their agreement on which pair of a random class triplet merges first. The configuration of metric and linkage is selected per dataset by a criterion fixed without reference to the agreements: the highest cophenetic fidelity among configurations whose reference cuts are non-degenerate for every model. Local agreement is read with the permutation calibration of Gröger et al. (2026), on mutual-kNN agreement and on centroid CKA, with permutations of the class correspondence as the null.

3.6 PROJECTION AND TASKS

Three readouts are compared on the intact representation. A parameter-free map centers the features, scales the 95th-percentile norm to radius $1/\sqrt{2}$ and preserves directions. Euclidean, cosine and Poincaré readouts are then compared on the same representation. Nearest-centroid classification and 5-way 5-shot episodes score the three readouts on every cell of the task grid.

4 EXPERIMENTAL SETUP

The census has twelve backbones, six class sets and fifteen text models. The vision backbones are four supervised ViTs trained on ImageNet-21k, DINO-B, four DINOv2 sizes, CLIP-B, CLIP-L and SigLIP-B. The class sets are ImageNet and CIFAR-100, which carry a real hierarchy, CIFAR-10 and DTD, and the flat FMNIST and MNIST. The text models read the thousand ImageNet class names: four GPT-2 sizes, three Pythia sizes, OLMo-1B and seven sentence embedders, which also read DBpedia Classes and HierarCaps. Table 5 lists the panel.

Four controls test the alternatives. Two self-supervised ResNet-50 backbones read the vision census. Two ViT-B trained on ImageNet-1k leaf labels, DeiT-B and the augreg ViT-B, test whether a hierarchical label set explains the depth. MERU and its Euclidean CLIP twin, in three sizes, test whether training in hyperbolic space creates depth. The census ViT-B fine-tuned with and without a hierarchical objective is a trained control.

The premise is read on images and hierarchy on centroids. The sample-level reading uses about a thousand stratified training images per cell on CIFAR-100 and DTD. Class centroids average a hundred cached training images per class on ImageNet and the full cached split elsewhere. Causal language models give the last-token state of “a photo of a {class}”, extracted one prompt at a time because batching alters GPT-2’s hidden states, and embedders use their recommended pooling (Table 6).

Every reading is calibrated against two hundred replicates. Each cell is read against two hundred Haar replicates, with half a million sampled quadruples per seed and ten seeds. With ten classes there are only 210 quadruples, so $\hat{\delta}_{99.9}$ coincides with the supremum there. Benjamini–Hochberg runs over the vision census and, separately, over the text census. The depth test uses the WordNet cut into thirty superclasses on ImageNet, the twenty coarse labels on CIFAR-100, a balanced frame as a control, and ten star seeds.

5 RESULTS

5.1 THE PREMISE AT THE SAMPLE LEVEL

Raw readings are not evidence, ours or published, and calibrated ones are weak and model-dependent. On per-image features, where the premise is read, the calibrated reading is not genuine in most cells. Where it is genuine, the excess removes at most two fifths of the null reading. Table 7 gives the reading per cell.

A published reading is reproduced and calibrated. Their estimator on our extraction reproduces the raw δ_{rel} that Khrulkov et al. (2020) report for ResNet-34 on four datasets within 0.03. Calibrated on the same clouds, the excess is negative on every dataset, and only CIFAR-100 and MiniImageNet fall below the null in every batch at the 0.05 level. Table 7 gives the four rows.

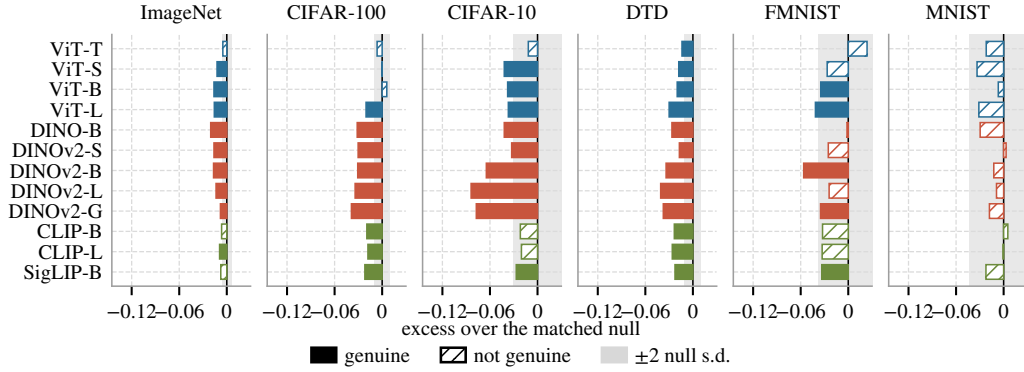


Figure 3: **Most cells show structure beyond the second moments.** Excess of the reading of record over its matched null, one panel per dataset, one bar per backbone, filled when genuine; band: two null standard deviations.

model	family	$\hat{\delta}_{99.9}$ IN	excess IN	$r/200$ (p)	exc. C100	exc. C10	ρ_{WN}
ViT-T	sup.	0.088	-0.005°	169 (.159)	-0.006°	-0.012°	+0.52
ViT-S	sup.	0.063	-0.012	200 (.005)	-0.000°	-0.042	+0.50
ViT-B	sup.	0.054	-0.016	200 (.005)	+0.006°	-0.038	+0.49
ViT-L	sup.	0.055	-0.016	200 (.005)	-0.020	-0.037	+0.53
DINO-B	SSL	0.055	-0.020	200 (.005)	-0.031	-0.042	+0.36
DINOv2-S	SSL	0.053	-0.016	200 (.005)	-0.030	-0.033	+0.22
DINOv2-B	SSL	0.034	-0.017	200 (.005)	-0.031	-0.065	+0.18
DINOv2-L	SSL	0.030	-0.014	200 (.005)	-0.034	-0.084	+0.18
DINOv2-G	SSL	0.033	-0.008	200 (.005)	-0.039	-0.077	+0.20
CLIP-B	contr.	0.084	-0.007°	172 (.144)	-0.019	-0.022°	+0.57
CLIP-L	contr.	0.076	-0.009	197 (.020)	-0.018	-0.020°	+0.59
SigLIP-B	contr.	0.075	-0.008°	189 (.060)	-0.022	-0.027	+0.58

Table 1: **Structure beyond the second moments is the rule at the class level.** Per backbone: raw ImageNet reading $\hat{\delta}_{99.9}$, excess over the Haar null mean, rank $r/200$ with left-tail p , genuine = BH $p \leq 0.05$ over 72 cells (°: not genuine; bold: above the null), the CIFAR-100 and CIFAR-10 excess, and ρ_{WN} , the Spearman correlation with WordNet distances.

5.2 STRUCTURE BEYOND THE SECOND MOMENTS AT THE CLASS LEVEL

Most cells show structure beyond the second moments. On class centroids the excess is negative in almost every cell. It is genuine in 44 of 72 cells and 30 of 36 on the three datasets with 47 classes or more. On ImageNet and CIFAR-100, the two sets with a real hierarchy, the count is 18 of 24. Figure 3 shows the census and Table 1 gives it row by row.

The count survives resampling. The census null accounts for replicate noise only. Folding image resampling and the estimator’s seeds into that spread, or repeating the census on resampled centroid sets, leaves most of the count in place. Table 2 gives the counts and the bootstrap.

No family owns the structure. Every family contributes genuine cells. The smallest backbone is genuine on one dataset only, and the flat datasets have smaller and less consistent excesses; FMNIST is genuine in 5 of 12 backbones. A class-count control shows that the excess shrinks with the number of classes, so only the verdict carries across datasets. Tables 4 and 2 give the details.

A star of clusters already passes the census. A star of clusters has structure beyond second moments, and a synthetic star with no hierarchy scores an excess as deep as any real cell. A negative excess therefore says structure beyond the second moments, not depth. Table 3 gives the star’s excess.

Neural collapse is the flat limit, not what the census sees. Neural collapse predicts that supervised class means converge to a simplex frame (Papayan et al., 2020), a star with no hierarchy. The census does not certify that frame: class means cluster around superclass hubs, which the superclass recovery of Section 5.4 shows and the depth test reads as hub alignment. The same structure appears without labels and in language models.

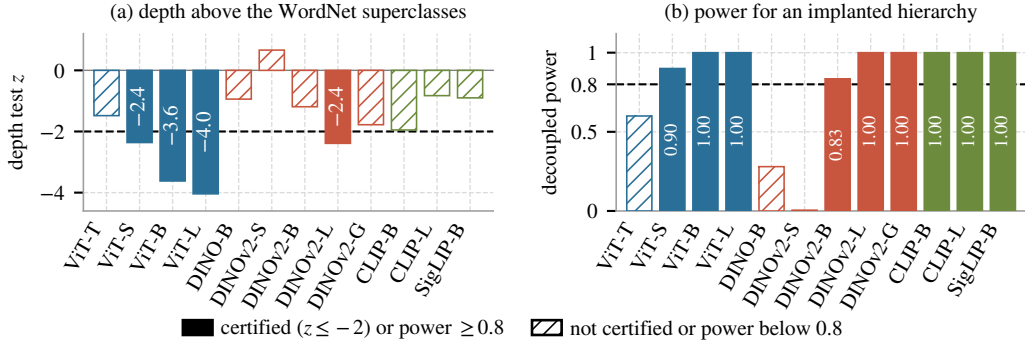


Figure 4: **The depth test certifies in 4 of 12 ImageNet backbones that clusters are oriented toward their hubs, not that the hubs form a hierarchy; no hub hierarchy is found in the nine backbones where the decoupled control detects an implanted one, and the test is blind in the other three.** (a) Depth z per backbone, filled when certified; line at $z = -2$. (b) Decoupled power per backbone for an implanted three-level hierarchy at the backbone’s own spectrum and noise level, filled at 0.8 or more; line at 0.8.

5.3 DEPTH ABOVE THE LABELLED CLUSTERS

The depth test certifies that clusters are oriented toward their hubs in four backbones. It certifies 4 of 12 backbones, ViT-S, ViT-B, ViT-L and DINOv2-L, under both matched stars. It never fires in the other direction. The noise level of a cloud is its within-cluster spread relative to the distance between its hubs. Figure 4a shows the z per backbone and Table 8 gives both stars.

No hub hierarchy is found where the control has power. The implant is a two-level tree over the thirty hubs, whereas WordNet has up to fourteen levels above them. A decoupling control keeps the real hubs and Haar-rotates each cluster’s offsets, randomizing orientations only. Once cluster orientations are randomized, none of the four fires, and ViT-L reads -1.61 , within the -1.61 to -1.76 of the flat control: without its orientations the real cloud reads like a flat one. The test misses an implanted two-level tree at this noise level but detects a three-level hierarchy with ViT-L’s spectrum and noise in 4 of 5 seeds, and that detection survives orientation randomization, whereas none of the four real verdicts does. Table 8 gives the decoupling control and Figure 6 the detection rates.

The power of the test follows the backbone, not its ratio or family. The decoupled control detects an implanted hierarchy in ViT-S/B/L, DINOv2-B/L/G, CLIP-B/L and SigLIP-B (decoupled power 0.83–1.00), while ViT-T, DINO-B and DINOv2-S are not covered (power 0.01–0.60), so no hub hierarchy is found in the nine and the test is blind in the other three. Its false alarms on the flat control are 8 of 50 decoupled runs on the frame of record. The three blind backbones sit at ratios 1.5 to 3.0, inside the 1.3 to 3.9 of the nine covered. Table 9 and Figure 4b give the power per backbone.

The alignment is not the spread of feature norms. The alignment survives removing the radial component of every offset in all four certified backbones, z from -2.26 to -3.99 under both stars, which excludes the spread of feature norms as its source. Under full L2 normalization, which also moves the hubs, it survives in ViT-B and ViT-L only. Table 8 gives the transformed readings.

A deep hierarchy at the real noise level is detected. A synthetic cloud with ViT-L’s spectrum and a three-level hierarchy implanted at ViT-L’s within/between ratio fires in 4 of 5 seeds at the frame of record. A flat control with the same spectrum and ratio fires in 0 of 5, and once decoupled it fires in 8 of 50 runs. The decoupled test is therefore biased toward firing, so the four real backbones not firing once decoupled is conservative evidence, and the synthetic hierarchy at $z = -4.19$ to -4.82 stands well clear of that bias, the flat control at -1.61 to -1.76 . The decoupling control fires on every deep seed and reads deeper than the intact cloud. The deeper reading comes from both sides: decoupling shrinks the star spread and deepens the excess below the star. The WordNet Poincaré embeddings (Nickel & Kiela, 2017), read with Euclidean distances, do not fire, and Table 9 gives every reading.

A trained hierarchy survives decoupling. ViT-B was fine-tuned on ImageNet-1k with leaf cross-entropy alone and with an added hierarchical cross-entropy at the WordNet cuts of thirty, six and two superclasses, one seed, identical batches. The hierarchical model is certified on both frames and keeps firing once decoupled, 10 of 10, whereas the leaf-CE and frozen models do not, 0 of 10, on the frame of record. On the balanced frame the frozen checkpoint also fires in 7 of 10. That frame’s

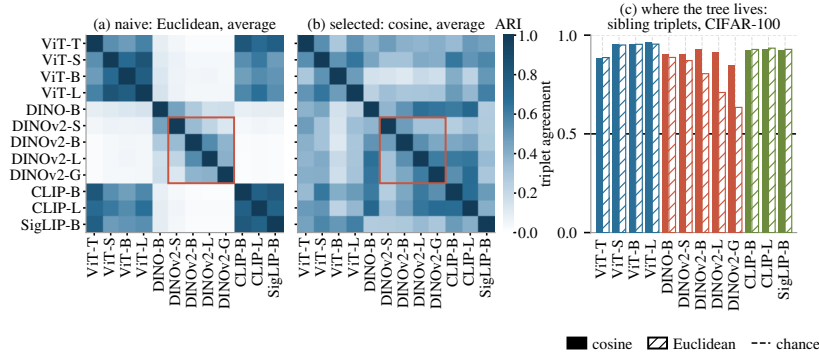


Figure 5: **The island is a chaining artifact; a moderate gap remains, and the self-supervised tree lives in angles.** (a) Pairwise ARI between dendrogram cuts under the naive configuration, where the DINOv2 block agrees with the rest at 0.03; (b) under the selected configuration, where it agrees at 0.38; (c) sibling-triplet agreement under cosine and Euclidean distance: DINOv2 loses most of its structure under Euclidean distance, the other families do not.

flat control fires once decoupled in 0 of 50 runs, so ViT-B shows hub structure under that grouping and the frame of record does not. The pre-set criterion asked the leaf-CE and frozen models not to be certified intact, which they are, by alignment, so the discriminating comparison is the decoupled one. Table 8 gives the readings.

The test never fires on randomized hubs. With the hubs of every real cloud randomized, the test fires in 0 of 60 runs, the four certified backbones included. The alignment is relational: it needs both the real hubs and the real orientation of each cluster relative to them. Randomizing either removes it: 0 of 60 with random hubs and 0 of 4 with random orientations. Table 9 gives the runs.

The certified set depends on the frame. Under a balanced frame of thirty superclasses ViT-T joins the certified set and ViT-S and DINOv2-L leave it, and cutting WordNet at ten or sixty superclasses changes the set again. Only ViT-B and ViT-L are certified under every frame. Table 8 gives the frames.

Leaf labels can produce the alignment but do not guarantee it. The avgreg ViT-B trained on ImageNet-1k leaf labels is certified and aligned with WordNet like the ImageNet-21k ViTs, whereas DeiT-B, trained on the same labels with another recipe, is neither. A hierarchical label set therefore does not explain the certified ViTs’ depth, and taxonomy alignment is recipe-dependent. Table 8 gives the two models.

Imposing the geometry does not create hub structure. MERU embeds images on the Lorentz hyperboloid with an entailment objective, and its Euclidean twin shares backbone, data and recipe (Desai et al., 2023). MERU shows the same clustering, no such structure either, and embeddings that never leave the near-flat regime, where Lorentz and Euclidean distances almost coincide. The control therefore tests the objective rather than a curved geometry. Table 8 gives the readings and the radii.

5.4 WHOSE TREE

The naive map manufactures an island. Under Euclidean distances with average linkage the DINOv2 family agrees with the supervised and contrastive block far less than the block agrees with itself. The cause is chaining: DINOv2’s Euclidean dendrograms put almost every class in one cluster, forcing the index to zero. Figure 5a shows the naive map.

Controlling the cut leaves a moderate gap. The criterion of Section 3.5 selects cosine distances with average linkage on ImageNet. Under it the island shrinks to a moderate gap: the sibling-triplet agreement of DINOv2 with the block is 0.77 against 0.76 within the block. About a third of the within-block agreement is missing under the admissible configurations as a whole. Figure 5b shows the selected map and Table 10 gives the six configurations.

The self-supervised tree is angular. DINOv2’s semantics concentrate in the angular component: for DINOv2-L and DINOv2-G, sibling triplets are resolved far better under cosine than under Euclidean distance, so the Euclidean dendrograms chain. The census repeated on cosine geometry agrees with the Euclidean census in most cells, and DINOv2 on ImageNet is genuine under both. Figure 5c shows the triplets and Table 4 gives the cosine census.

Alignment with WordNet follows supervision and recipe. The correlation between inter-centroid and WordNet distances is highest for the contrastive VLMs, then the supervised ViTs,

and lowest for DINOv2, with shuffle and Gaussian controls at zero. Supervision deepens the resemblance on average, and DeiT-B shows that the effect is recipe-dependent. Table 11 gives the correlations.

The recovery is not WordNet circularity. On DBpedia Classes, an ontology independent of WordNet, the three text embedders read a genuine excess and recover the true partition under every configuration. On HierarCaps, a four-level caption hierarchy with no class set (Alper & Averbuch-Elor, 2024), the encoders resolve sibling triplets far above chance. Table 11 gives both.

Text depends on recipe and scale. On the ImageNet class names, 7 of 15 text models are genuine under the record. GPT-2 S is genuine at the margin, GPT-2 M is not, and GPT-2 L and XL are genuine under every reading, as are Pythia at every scale and OLMo-1B. Table 6 gives the census.

5.5 WHAT IS SHARED IS LOCAL

Models share neighborhoods, not metrics. Permutation-calibrated mutual-kNN agreement and centroid CKA survive across all sixty-six model pairs. The Poincaré readout improves neither: models share who is near whom and differ in the metric (Gröger et al., 2026). Table 12 gives the values.

6 IMPLICATIONS FOR HYPERBOLIC REPRESENTATION LEARNING

The raw reading cannot select a curvature. Khrulkov et al. (2020) set the curvature of the Poincaré ball from the relative hyperbolicity, $c = (0.144/\delta_{\text{rel}})^2$ with $\delta_{\text{rel}} = 2\delta/\text{diam}$. The rule, calibrated with the supremum, assigns structureless clouds on the Gaussian band, itself the sampled supremum rather than the census percentile, a curvature from 0.48 to 2.5, tracking the width of the backbone. Table 3 gives the band.

Both readings predict the gain, and the depth verdict predicts none. Within each hierarchical dataset the raw reading predicts the Poincaré readout’s gain as well as the calibrated excess does. The depth verdict predicts no hyperbolic gain. Calibration buys interpretation, a low δ is not hierarchy, and not prediction. Table 13 gives both predictions and the policies.

The calibration certifies structure and does not choose the readout. Before imposing curvature, certify that there is structure to justify it, then take the cheapest readout that collects it. Cosine collects the self-supervised structure, and the Poincaré readout adds +0.9 to +1.3 pp over cosine for the contrastive VLMs on prototype tasks and nothing consistent elsewhere. In few-shot learning a fixed-radius Euclidean encoder does at least as well as hyperbolic prototypes (Moreira et al., 2024). Table 13 gives the gains.

7 CONCLUSION AND LIMITATIONS

Conclusion. Read correctly, foundation models organize classes into clustered structure that is moderately shared and does not converge to one common tree; no hub hierarchy is found where the test has power; and their raw tree-likeness is not evidence for hyperbolic geometry. Curvature should be measured rather than assumed. A change of readout collects what is there.

Limitations of the instrument. (i) The excess certifies structure beyond second moments, star-like clustering included, so depth needs the separate test. (ii) The spectrum null conditions on the second moments, to which a hierarchy contributes, so a low excess does not exclude hierarchy carried by the spectrum. (iii) The depth test’s power for a three-level hierarchy follows the backbone rather than its within/between ratio or family. Under the decoupled control it is 0.83 or more in nine backbones and 0.60 or less in the other three. The verdict of no hub hierarchy therefore holds in the nine only, at 8 of 50 false alarms on flat clouds. It misses a two-level implant at this noise level, and the decoupled test is biased toward firing, which makes the real verdicts under decoupling conservative. It certifies alignment above its frame only, and its trained positive control, one seed, discriminates the objectives only once decoupled. The alignment is not the radial spread of feature norms, which a control removes without changing the verdicts; full L2 normalization, a stronger transformation, keeps it in two of the four backbones. Implanted alignment is detected in 60% of runs at full strength, in every seed for the supervised ViTs and CLIP-B and in none for the DINOv2 family, so the power of the test for alignment is backbone-dependent and the DINOv2-L certification rests on a structure the implant does not reproduce. (iv) The depth test is Euclidean, and for the angular DINOv2 tree it may be conservative. (v) A causal LM’s class representation depends on the extraction protocol as much as on the model.

Limitations of scope. (vi) The census is class-centroid geometry on six datasets, and the sample-level reading covers two datasets, one image budget and one published setting. (vii) The objective-geometry link is correlational, resting chiefly on DINOv2’s scale range and on two leaf-label ViTs. (viii) Downstream scope is prototype-based metric tasks with modest gains.

REPRODUCIBILITY STATEMENT

Every number in the paper is annotated in the source with the result file that produces it, and all tables are generated by script from those files (Table 14). The full pipeline with fixed seeds will be released in the anonymized repository (`TODO(author)`). Its `tool/` directory ships the instrument as one script that reproduces any cell of Table 1 from its centroid matrix. Estimator settings are stated in Section 4 and the model panel in Table 5.

ETHICS STATEMENT

This work analyzes publicly released models and standard public datasets, involves no human subjects and no personal data, and proposes an analysis methodology and a metric-selection heuristic. We do not foresee ethical risks beyond those already associated with the underlying models and datasets.

AI USE STATEMENT

In this work, we used generative AI tools for writing assistance; for the retrieval of references; and for research ideation or execution, namely the implementation and execution of experiments under the authors’ direction and LLM-simulated reviews used as methodological feedback. We have not used generative AI tools for other tasks with required disclosure. We have reviewed all AI-assisted work, and we take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

REFERENCES

- Aaron B. Adcock, Blair D. Sullivan, and Michael W. Mahoney. Tree-like structure in large social and information networks. In *IEEE International Conference on Data Mining (ICDM)*, pp. 1–10, 2013.
- Charu C. Aggarwal, Alexander Hinneburg, and Daniel A. Keim. On the surprising behavior of distance metrics in high dimensional space. In *Database Theory – ICDT 2001*, volume 1973 of *Lecture Notes in Computer Science*, pp. 420–434. Springer, 2001.
- Morris Alper and Hadar Averbuch-Elor. Emergent visual-semantic hierarchies in image-text representations. In *European Conference on Computer Vision (ECCV)*, 2024.
- Alessio Ansuini, Alessandro Laio, Jakob H. Macke, and Davide Zoccolan. Intrinsic dimension of data representations in deep neural networks. In Hanna M. Wallach, Hugo Larochelle, Alina Beygelzimer, Florence d’Alché-Buc, Emily B. Fox, and Roman Garnett (eds.), *Advances in Neural Information Processing Systems 32: Annual Conference on Neural Information Processing Systems 2019, NeurIPS 2019, December 8-14, 2019, Vancouver, BC, Canada*, pp. 6109–6119, 2019. URL <https://proceedings.neurips.cc/paper/2019/hash/cfcce0621b49c983991ead4c3d4d3b6b-Abstract.html>.
- Mina Ghadimi Atigh, Julian Schoep, Erman Acar, Nanne van Noord, and Pascal Mettes. Hyperbolic image segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 4453–4462, June 2022.
- Ahmad Bdeir, Kristian Schwethelm, and Niels Landwehr. Fully hyperbolic convolutional neural networks for computer vision. In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net, 2024. URL <https://openreview.net/forum?id=ekz1hN5QNh>.
- Kevin Beyer, Jonathan Goldstein, Raghu Ramakrishnan, and Uri Shaft. When is “nearest neighbor” meaningful? In *Database Theory – ICDT’99*, volume 1540 of *Lecture Notes in Computer Science*, pp. 217–235. Springer, 1999.
- Ines Chami, Zhitao Ying, Christopher Ré, and Jure Leskovec. Hyperbolic graph convolutional neural networks. In Hanna M. Wallach, Hugo Larochelle, Alina Beygelzimer, Florence d’Alché-Buc, Emily B. Fox, and Roman Garnett (eds.), *Advances in Neural Information Processing Systems 32: Annual Conference on Neural Information Processing Systems 2019, NeurIPS 2019, December 8-14, 2019, Vancouver, BC, Canada*, pp. 4869–4880, 2019.

- Weize Chen, Xu Han, Yankai Lin, Hexu Zhao, Zhiyuan Liu, Peng Li, Maosong Sun, and Jie Zhou. Fully hyperbolic neural networks. In Smaranda Muresan, Preslav Nakov, and Aline Villavicencio (eds.), *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 5672–5686, Dublin, Ireland, May 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.acl-long.389. URL <https://aclanthology.org/2022.acl-long.389/>.
- Karan Desai, Maximilian Nickel, Tanmay Rajpurohit, Justin Johnson, and Ramakrishna Vedantam. Hyperbolic image-text representations. In *Proceedings of the 40th International Conference on Machine Learning, ICML’23*. JMLR.org, 2023.
- Aleksandr Ermolov, Leyla Mirvakhabova, Valentin Khrulkov, Nicu Sebe, and Ivan V. Oseledets. Hyperbolic vision transformers: Combining improvements in metric learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2022, New Orleans, LA, USA, June 18-24, 2022*, pp. 7399–7409. IEEE, 2022. doi: 10.1109/CVPR52688.2022.00726. URL <https://doi.org/10.1109/CVPR52688.2022.00726>.
- Hervé Fournier, Anas Ismail, and Antoine Vigneron. Computing the gromov hyperbolicity of a discrete metric space. *Information Processing Letters*, 115(6-8), 2015.
- Octavian Ganea, Gary Becigneul, and Thomas Hofmann. Hyperbolic neural networks. In S. Bengio, H. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, and R. Garnett (eds.), *Advances in Neural Information Processing Systems*, volume 31. Curran Associates, Inc., 2018.
- Fabian Gröger, Song Wen, and Maria Brbić. Revisiting the platonic representation hypothesis: An aristotelian view. In *International Conference on Machine Learning (ICML)*, 2026. arXiv:2602.14486.
- M. Gromov. Hyperbolic groups. In S. M. Gersten (ed.), *Essays in Group Theory*, pp. 75–263. Springer New York, New York, NY, 1987. ISBN 978-1-4613-9586-7. doi: 10.1007/978-1-4613-9586-7_3. URL https://doi.org/10.1007/978-1-4613-9586-7_3.
- Albert Gu, Frederic Sala, Beliz Gunel, and Christopher Ré. Learning mixed-curvature representations in product spaces. In *International Conference on Learning Representations*, 2019.
- Neil He, Rishabh Anand, Hiren Madhu, Ali Maatouk, Smita Krishnaswamy, Leandros Tassiulas, Menglin Yang, and Rex Ying. HELM: Hyperbolic large language models via mixture-of-curvature experts. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025. URL <https://openreview.net/forum?id=RnbJPkakkM>.
- Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. Position: The platonic representation hypothesis. In Ruslan Salakhutdinov, Zico Kolter, Katherine A. Heller, Adrian Weller, Nuria Oliver, Jonathan Scarlett, and Felix Berkenkamp (eds.), *Forty-first International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21-27, 2024*, Proceedings of Machine Learning Research, pp. 20617–20642. PMLR / OpenReview.net, 2024. URL <https://proceedings.mlr.press/v235/huh24a.html>.
- W. Sean Kennedy, Onuttom Narayan, and Iraj Saniee. On the hyperbolicity of large-scale networks. *arXiv preprint arXiv:1307.0031*, 2013.
- Valentin Khrulkov, Leyla Mirvakhabova, Evgeniya Ustinova, Ivan Oseledets, and Victor Lempitsky. Hyperbolic image embeddings. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2020.
- A. Sophia Koepke, Daniil Zverev, Shiry Ginosar, and Alexei A. Efros. Back into plato’s cave: Examining cross-modal representational convergence at scale, 2026. arXiv preprint.
- Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey E. Hinton. Similarity of neural network representations revisited. In Kamalika Chaudhuri and Ruslan Salakhutdinov (eds.), *Proceedings of the 36th International Conference on Machine Learning, ICML 2019, 9-15 June 2019, Long Beach, California, USA*, Proceedings of Machine Learning Research, pp. 3519–3529. PMLR, 2019. URL <http://proceedings.mlr.press/v97/kornblith19a.html>.

- Gabriel Moreira, Manuel Marques, João Paulo Costeira, and Alexander Hauptmann. Hyperbolic vs Euclidean embeddings in few-shot learning: Two sides of the same coin. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pp. 2082–2090, 2024.
- Onuttom Narayan and Iraj Saniee. Large-scale curvature of networks. *Physical Review E*, 84(6): 066108, 2011.
- Bogdan Nica and Ján Špakula. Strong hyperbolicity. *Groups, Geometry, and Dynamics*, 10(3), 2016. arXiv:1408.0250.
- Maximilian Nickel and Douwe Kiela. Poincaré embeddings for learning hierarchical representations. In Isabelle Guyon, Ulrike von Luxburg, Samy Bengio, Hanna M. Wallach, Rob Fergus, S. V. N. Vishwanathan, and Roman Garnett (eds.), *Advances in Neural Information Processing Systems 30: Annual Conference on Neural Information Processing Systems 2017, December 4-9, 2017, Long Beach, CA, USA*, pp. 6338–6347, 2017. URL <https://proceedings.neurips.cc/paper/2017/hash/59dfa2df42d9e3d41f5b02bfc32229dd-Abstract.html>.
- Avik Pal, Max van Spengler, Guido Maria D’Amely di Melendugno, Alessandro Flaborea, Fabio Galasso, and Pascal Mettes. Compositional entailment learning for hyperbolic vision-language models. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=3il3Gev2hV>.
- Vardan Papyan, X. Y. Han, and David L. Donoho. Prevalence of neural collapse during the terminal phase of deep learning training. *Proceedings of the National Academy of Sciences*, 117(40): 24652–24663, 2020. doi: 10.1073/pnas.2015509117. URL <https://www.pnas.org/doi/abs/10.1073/pnas.2015509117>.
- Kiho Park, Yo Joong Choe, Yibo Jiang, and Victor Veitch. The geometry of categorical and hierarchical concepts in large language models. *arXiv preprint arXiv:2406.01506*, 2024.
- Phillip Pope, Chen Zhu, Ahmed Abdelkader, Micah Goldblum, and Tom Goldstein. The intrinsic dimension of images and its impact on learning. In *9th International Conference on Learning Representations, ICLR 2021, Virtual Event, Austria, May 3-7, 2021*. OpenReview.net, 2021. URL <https://openreview.net/forum?id=XJk19XzGq2J>.
- Frederic Sala, Christopher De Sa, Albert Gu, and Christopher Ré. Representation tradeoffs for hyperbolic embeddings. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pp. 4460–4469. PMLR, 2018.
- Rik Sarkar. Low distortion delaunay embedding of trees in hyperbolic plane. In Marc J. van Kreveld and Bettina Speckmann (eds.), *Graph Drawing - 19th International Symposium, GD 2011, Eindhoven, The Netherlands, September 21-23, 2011, Revised Selected Papers*, Lecture Notes in Computer Science, pp. 355–366. Springer, 2011. doi: 10.1007/978-3-642-25878-7_34. URL https://doi.org/10.1007/978-3-642-25878-7_34.
- Aditya Sinha, Siqi Zeng, Makoto Yamada, and Han Zhao. Learning structured representations with hyperbolic embeddings. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang (eds.), *Advances in Neural Information Processing Systems*, volume 37, pp. 91220–91259. Curran Associates, Inc., 2024. doi: 10.52202/079017-2895.
- Alexandru Tifrea, Gary Bécigneul, and Octavian-Eugen Ganea. Poincaré glove: Hyperbolic word embeddings. In *International Conference on Learning Representations (ICLR)*, 2019.
- Menglin Yang, Ram Samarth B B, Aosong Feng, Bo Xiong, Jiahong Liu, Irwin King, and Rex Ying. Hyperbolic fine-tuning for large language models. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025. URL <https://openreview.net/forum?id=TkEdQv0bXB>.

A PROOFS

Proof of Proposition 1. (a) Each pairing sum in Eq. 1 adds two distances of the space, each of which lies in $[d_{\min}, d_{\max}]$. Hence $S_{\max} \leq 2d_{\max}$ and $S_{\min} \geq 2d_{\min}$, so the defect $\frac{1}{2}(S_{\max} - S_{\min})$ is at most $d_{\max} - d_{\min}$ for every quadruple, and so is its supremum δ . Dividing by the diameter $d_{\max}$ gives $\delta_{\text{norm}} \leq 1 - d_{\min}/d_{\max}$; the same bound holds for the 99.9th percentile of the sampled defects, which never exceeds their supremum. (b) Let $x_1, \dots, x_n$ be iid standard Gaussian in $\mathbb{R}^d$ with n fixed. For any pair $i \neq j$ the coordinates of $x_i - x_j$ are iid with variance 2, so $\|x_i - x_j\|^2/(2d) \rightarrow 1$ almost surely as $d \rightarrow \infty$ by the law of large numbers, and the convergence holds jointly over the finitely many pairs. Therefore $d_{\min}/d_{\max} \rightarrow 1$, and by part (a) $\delta_{\text{norm}} \rightarrow 0$. The rate is the distance-concentration rate of Beyer et al. (1999) and Aggarwal et al. (2001): the relative contrast $(d_{\max} - d_{\min})/d_{\min}$ vanishes as $d^{-1/2}$. Part (b) is the dimension confound alone. The statistic confound is separate and elementary: a maximum over a growing sample of quadruples is non-decreasing in the sample size, so the sampled supremum can only rise with the budget. $\square$

B ONE TABLE PER QUESTION

Every table is generated by script from the released per-cell result files, and no value is hand-transcribed; the tables are numbered in the order the main text first cites them, one per question, with their result files indexed in Table 14.

B.1 IS THE READING ROBUST? QUADRUPLE BUDGET, NULL CONSTRUCTION, RESAMPLING AND CLASS COUNT

(a) Quadruple budget under the record (Haar null, p99.9, 200 replicates): the excess of nine cells at 10^4 to 2×10^6 sampled quadruples per seed; drift = range of the excess over budgets $\geq 10^5$, also divided by the excess's null spread.

model	data	10^4	5×10^4	10^5	5×10^5	10^6	2×10^6	drift ($\geq 10^5$)	drift/s.d.
ViT-L	IN	-0.0161	-0.0161	-0.0160	-0.0157	-0.0159	-0.0159	0.0002	0.07
DINOv2-L	IN	-0.0131	-0.0136	-0.0137	-0.0135	-0.0136	-0.0136	0.0002	0.38
CLIP-B	IN	-0.0063	-0.0060	-0.0059	-0.0065	-0.0064	-0.0064	0.0007	0.13
ViT-L	C100	-0.0208	-0.0208	-0.0206	-0.0205	-0.0205	-0.0205	0.0002	0.03
DINOv2-L	C100	-0.0348	-0.0348	-0.0348	-0.0348	-0.0347	-0.0347	0.0001	0.07
CLIP-B	C100	-0.0186	-0.0195	-0.0188	-0.0195	-0.0194	-0.0194	0.0006	0.10
ViT-L	DTD	-0.0305	-0.0308	-0.0312	-0.0311	-0.0313	-0.0313	0.0002	0.05
DINOv2-L	DTD	-0.0418	-0.0419	-0.0421	-0.0420	-0.0422	-0.0420	0.0002	0.06
CLIP-B	DTD	-0.0243	-0.0235	-0.0242	-0.0240	-0.0243	-0.0242	0.0003	0.05

Table 2: **The excess is budget-stable, the genuine count survives resampling and estimator noise, the excess shrinks with the class count, and training moves it.** (a) Excess against the quadruple budget for nine cells; drift = range of the record excess over budgets $\geq 10^5$, divided by the excess's spread in the last column; under the record every sign holds at every budget. (b) Bootstrap of the excess over resampled images per class on ImageNet, CIFAR-100 and DTD: the bootstrap s.d. is at most 0.0026 and every cell keeps the sign of its record excess in at least 0.67 of its resamples.

(b) Centroid bootstrap under the record (Haar null, p99.9): 30 resamples with replacement of the cached training images per class at the census budget; ImageNet with 20 null replicates per resample, CIFAR-100 and DTD with 200.

model	dataset	images/class	null rep.	excess (reference)	bootstrap mean	bootstrap s.d.	fraction negative
ViT-T	ImageNet	100	20	-0.0055	-0.0051	0.0011	1.00
ViT-S	ImageNet	100	20	-0.0126	-0.0122	0.0007	1.00
ViT-B	ImageNet	100	20	-0.0165	-0.0160	0.0006	1.00
ViT-L	ImageNet	100	20	-0.0160	-0.0160	0.0004	1.00
DINO-B	ImageNet	100	20	-0.0209	-0.0208	0.0003	1.00
DINOv2-S	ImageNet	100	20	-0.0164	-0.0162	0.0004	1.00
DINOv2-B	ImageNet	100	20	-0.0167	-0.0164	0.0002	1.00
DINOv2-L	ImageNet	100	20	-0.0136	-0.0134	0.0001	1.00
DINOv2-G	ImageNet	100	20	-0.0080	-0.0083	0.0004	1.00
CLIP-B	ImageNet	100	20	-0.0068	-0.0069	0.0009	1.00
CLIP-L	ImageNet	100	20	-0.0094	-0.0095	0.0006	1.00
SigLIP-B	ImageNet	100	20	-0.0080	-0.0081	0.0006	1.00
ViT-T	CIFAR-100	500	200	-0.0064	-0.0059	0.0018	1.00
ViT-S	CIFAR-100	500	200	-0.0003	-0.0004	0.0012	0.67
ViT-B	CIFAR-100	500	200	+0.0057	+0.0055	0.0011	0.00
ViT-L	CIFAR-100	500	200	-0.0205	-0.0205	0.0006	1.00
DINO-B	CIFAR-100	500	200	-0.0316	-0.0315	0.0006	1.00
DINOv2-S	CIFAR-100	500	200	-0.0306	-0.0306	0.0004	1.00
DINOv2-B	CIFAR-100	500	200	-0.0313	-0.0313	0.0004	1.00
DINOv2-L	CIFAR-100	500	200	-0.0348	-0.0348	0.0003	1.00
DINOv2-G	CIFAR-100	500	200	-0.0395	-0.0395	0.0002	1.00
CLIP-B	CIFAR-100	500	200	-0.0195	-0.0192	0.0007	1.00
CLIP-L	CIFAR-100	500	200	-0.0181	-0.0183	0.0009	1.00
SigLIP-B	CIFAR-100	500	200	-0.0220	-0.0218	0.0008	1.00
ViT-T	DTD	80	200	-0.0142	-0.0141	0.0016	1.00
ViT-S	DTD	80	200	-0.0186	-0.0189	0.0020	1.00
ViT-B	DTD	80	200	-0.0208	-0.0212	0.0017	1.00
ViT-L	DTD	80	200	-0.0311	-0.0302	0.0017	1.00
DINO-B	DTD	80	200	-0.0272	-0.0267	0.0024	1.00
DINOv2-S	DTD	80	200	-0.0179	-0.0171	0.0025	1.00
DINOv2-B	DTD	80	200	-0.0350	-0.0345	0.0016	1.00
DINOv2-L	DTD	80	200	-0.0420	-0.0417	0.0016	1.00
DINOv2-G	DTD	80	200	-0.0386	-0.0387	0.0026	1.00
CLIP-B	DTD	80	200	-0.0240	-0.0242	0.0021	1.00
CLIP-L	DTD	80	200	-0.0266	-0.0270	0.0022	1.00
SigLIP-B	DTD	80	200	-0.0233	-0.0234	0.0014	1.00

Table 2: (continued)

(c) Joint sensitivity of the genuine count: z_{joint} against null, bootstrap and estimator noise, and (in parentheses) the number of the 30 centroid resamples in which the cell is genuine.

model	IN	C100	C10	DTD	FMNIST	MNIST
ViT-T	-1.1 (0) ^o	-0.7 (0) ^o	-0.6 (0) ^o	-1.9 (12) [†]	+1.3 (0) ^o	-0.9 (0) ^o
ViT-S	-3.6 (30)	-0.0 (0) ^o	-2.4 (29)	-3.0 (30)	-1.5 (30) ^{o‡}	-1.3 (0) ^o
ViT-B	-5.2 (30)	+0.8 (0) ^o	-2.1 (1) [†]	-3.9 (30)	-1.9 (30) [†]	-0.3 (0) ^o
ViT-L	-5.0 (30)	-4.0 (30)	-2.4 (30)	-6.4 (30)	-2.2 (30)	-1.4 (0) ^o
DINO-B	-8.9 (30)	-7.6 (30)	-2.4 (30)	-4.4 (30)	-0.1 (0) ^o	-1.4 (0) ^o
DINOv2-S	-15.9 (30)	-9.7 (30)	-2.2 (30)	-3.1 (30)	-1.3 (0) ^o	+0.2 (0) ^o
DINOv2-B	-31.7 (30)	-14.7 (30)	-4.9 (30)	-8.0 (30)	-2.6 (30)	-0.5 (0) ^o
DINOv2-L	-30.7 (30)	-21.6 (30)	-6.0 (30)	-10.6 (30)	-1.3 (30) ^{o‡}	-0.4 (0) ^o
DINOv2-G	-11.5 (30)	-28.2 (30)	-6.5 (30)	-8.2 (30)	-1.8 (23) [†]	-0.7 (0) ^o
CLIP-B	-1.3 (0) ^o	-3.1 (30)	-1.3 (0) ^o	-3.5 (30)	-1.4 (0) ^o	+0.3 (0) ^o
CLIP-L	-2.1 (11) [†]	-3.5 (30)	-1.3 (0) ^o	-4.2 (30)	-1.5 (0) ^o	-0.0 (0) ^o
SigLIP-B	-1.7 (0) ^o	-4.3 (30)	-1.7 (29) [†]	-4.0 (30)	-1.8 (30) [†]	-1.3 (0) ^o

Table 2: (continued) (c) Per cell: $z_{\text{joint}} = \text{excess} / \sqrt{\sigma_{\text{null}}^2 + \sigma_{\text{boot}}^2 + \sigma_{\text{est}}^2}$ and the number of resamples in which the cell is genuine under BH over the 72 cells; ^o: not genuine in the record; [†]: record-genuine but failing one of the two criteria ($z_{\text{joint}} \leq -2$, genuine in ≥ 27 of 30); [‡]: not record-genuine but passing one. Counts: record 44/72 (18/24); $z_{\text{joint}} \leq -2$: 39/72 (18/24); genuine in ≥ 27 of 30 resamples: 42/72 (17/24). (d) Class-count control under the record (means over subset seeds): the excess shrinks with the number of classes for coherent and random subsets alike, so magnitudes are not comparable across class counts and the verdict is what carries across datasets; “below” = subset seeds whose real value lies below the null at uncorrected $p \leq 0.05$; NC adv = prototype-classifier advantage of the Poincaré readout in pp. (e) Raw δ within architecture: fine-tuning on a task without class hierarchy raises it and it falls across transformer depth.

(d) Class count: ImageNet subsets of C classes, random (spanning the hierarchy) or WordNet-coherent (siblings, effectively flat), means over the subset seeds. The excess shrinks with the number of classes for coherent and random subsets alike, so magnitudes are not comparable across class counts; the verdict is what carries across datasets.

model	C	random subsets (span the hierarchy)				WordNet-coherent (siblings)			
		$\hat{\delta}_{99.9}$	excess	below	NC adv	$\hat{\delta}_{99.9}$	excess	below	NC adv
DINOv2-L	10	0.030	-0.134	5/5	+1.04	0.068	-0.098	5/5	+0.08
DINOv2-L	20	0.034	-0.091	5/5	+1.32	0.071	-0.065	5/5	+0.12
DINOv2-L	50	0.032	-0.046	5/5	+1.58	0.050	-0.042	5/5	+0.53
DINOv2-L	100	0.031	-0.032	5/5	+1.46	0.044	-0.031	5/5	+0.77
DINOv2-L	200	0.031	-0.022	5/5	+1.45	0.041	-0.023	5/5	+0.59
DINOv2-L	500	0.030	-0.016	2/2	+1.01	0.031	-0.018	2/2	+0.82
DINOv2-L	1000	0.030	-0.014	1/1	+0.57	—	—	—	—

Table 2: (continued)

(e) Training moves the raw reading within an architecture: one row per intervention. Statistic: the raw supremum δ_{norm} , the largest four-point defect over sampled quadruples divided by the diameter, on 512 sampled points; fine-tuning row: 1000 CIFAR-100 test images before and after a non-hierarchical (colour) fine-tuning of DINOv2-S, 5×10^4 quadruples; depth row: DINOv2-B hidden states per layer.

intervention	model	δ before	δ after	change
non-hierarchical fine-tuning	DINOv2-S	0.095	0.161	+69%
across depth (layer 1 $\rightarrow$ 12)	DINOv2-B	0.147	0.058	-61%

Table 2: (continued)

B.2 IS A LOW RAW READING EVIDENCE? CALIBRATION ON REFERENCE GEOMETRIES AND SYNTHETIC CLUSTERS

(a) Reference geometries.

space	absolute δ	δ_{norm}
balanced binary tree (depth 10)	0.000	0.000
$\mathbb{H}^2$ ($K = -1$), region radius $R=2/4/8/16$	0.65/0.69/0.69/0.69	0.162/0.087/0.043/0.022
uniform S^{99} (chord / geodesic)	0.25/0.37	0.143/0.179
iid Gaussian ($n=1000$), $d=192, \dots, 1536$	—	0.104/0.081/0.074/0.061/0.055/0.046

Table 3: **A low raw δ is not evidence of hierarchy, and the spectrum excess certifies structure beyond the second moments, not depth.** (a) Calibration on reference geometries: the estimator recovers the tree zero and the four-point constant of $\mathbb{H}^2$, spheres stay high, and iid Gaussians drift low as d grows, the dimension confound; the ordering trees $<$ hyperbolic $<$ spherical holds for hyperbolic regions of radius $R \geq 4$. Gaussian row: the sampled supremum from the released control file, the statistic the curvature rule uses; tree/ $\mathbb{H}^2$ /sphere rows re-verified synthetically at every freeze. (b) Excess of synthetic clouds (means over seeds; within/between noise ratio in parentheses) under the spectrum null A, with Gaussian coefficients and with Haar-rotated coefficients (exact sample spectrum), and under the hub-randomizing null B, which keeps every cluster and resamples the hub configuration. A pure star sits far below null A under either construction, so null A certifies structure beyond the second moments, not depth; null B also reports the star as hierarchical, since a near-regular simplex of hubs already minimizes δ , which is why the depth test of Table 8 is read against a matched star rather than against null B. Under the Haar construction the two-level hierarchy sits $2\text{--}3\times$ below the star; under the plain Gaussian null the mid-radius star scores -0.104 and the 6×5 hierarchy -0.14 , a fraction of either’s excess.

(b) Synthetic clusters: what each null certifies.

synthetic cloud ($n=1000$, $d=768$)	$\hat{\delta}$	spectrum null A		hub-randomizing null B	
		Gaussian	Haar	Gaussian	Haar
30-cluster star, tight (0.1)	0.055	-0.115	-0.115	-0.126	-0.058
30-cluster star, mid (0.3)	0.059	-0.104	-0.103	-0.114	-0.049
30-cluster star, loose (0.6)	0.061	-0.084	-0.082	-0.090	-0.033
2-level hierarchy 6×5 (0.3)	0.044	-0.143	-0.143	-0.113	-0.112

Table 3: (continued)

B.3 IS THE STRUCTURE BEYOND THE SECOND MOMENTS GENUINE? THE VISION CENSUS UNDER EVERY CONSTRUCTION

What each null preserves. The spectrum-matched null, Gaussian or Haar, preserves n , d , the mean and the covariance spectrum and destroys all higher-order structure; the PC-permutation null preserves each principal component’s marginal and destroys the dependence across components; the label shuffle keeps the point cloud itself and destroys the embedding–label correspondence; and the hub-randomizing null keeps each cluster’s geometry and hub assignment and destroys the hub arrangement, which is why it serves the depth test only when read against a matched star. “Genuine” in the text always means beyond the first of these. In plain terms, the spectrum null is a cloud with the same shape as the real one and no structure inside it; the hub null of Section 5 keeps every cluster of the real cloud and only rearranges where the clusters sit.

(a) The census of record per cell: Haar spectrum-matched null with Q orthogonal to the all-ones vector, 99.9th-percentile statistic, 200 replicates.

model	IN			C100			C10			DTD			FMNIST			MNIST		
	exc.	r	p	exc.	r	p	exc.	r	p	exc.	r	p	exc.	r	p	exc.	r	p
ViT-T	−0.005°	169	.159	−0.006°	151	.249	−0.012°	148	.264	−0.014	195	.030	+0.024°	19	.905	−0.022°	168	.164
ViT-S	−0.012	200	.005	−0.000°	112	.443	−0.042	199	.010	−0.018	199	.010	−0.027°	192	.045	−0.034°	183	.090
ViT-B	−0.016	200	.005	+0.006°	42	.791	−0.038	196	.025	−0.020	200	.005	−0.035	198	.015	−0.007°	122	.393
ViT-L	−0.016	200	.005	−0.020	200	.005	−0.037	200	.005	−0.030	200	.005	−0.041	200	.005	−0.031°	187	.070
DINO-B	−0.020	200	.005	−0.031	200	.005	−0.042	199	.010	−0.026	200	.005	−0.002°	104	.483	−0.030°	182	.095
DINOv2-S	−0.016	200	.005	−0.030	200	.005	−0.033	200	.005	−0.017	200	.005	−0.025°	184	.085	+0.003°	79	.607
DINOv2-B	−0.017	200	.005	−0.031	200	.005	−0.065	200	.005	−0.034	200	.005	−0.056	199	.010	−0.012°	142	.294
DINOv2-L	−0.014	200	.005	−0.034	200	.005	−0.084	200	.005	−0.041	200	.005	−0.024°	186	.075	−0.009°	135	.328
DINOv2-G	−0.008	200	.005	−0.039	200	.005	−0.077	200	.005	−0.037	200	.005	−0.035	197	.020	−0.018°	163	.189
CLIP-B	−0.007°	172	.144	−0.019	200	.005	−0.022°	184	.085	−0.023	200	.005	−0.032°	185	.080	+0.005°	81	.597
CLIP-L	−0.009	197	.020	−0.018	200	.005	−0.020°	180	.104	−0.026	200	.005	−0.033°	186	.075	−0.001°	97	.517
SigLIP-B	−0.008°	189	.060	−0.022	200	.005	−0.027	197	.020	−0.023	200	.005	−0.033	195	.030	−0.022°	176	.124

Table 4: **Structure beyond the second moments is the rule at the class level under every construction of the null and statistic, and the verdict does not depend on the metric.** (a) Per cell: excess of $\hat{\delta}_{99.9}$ over the null mean, r = replicates above the real value, $p = (1 + \#\{\text{null} \leq \text{real}\})/201$; genuine = Benjamini–Hochberg-corrected $p \leq 0.05$ over the 72 cells (°: not genuine). Genuine cells: 44/72 overall, 18/24 on ImageNet+CIFAR-100; sign agreement with the original 5-replicate Gaussian $\times$ supremum census on the 72 comparable cells: 65/72. (b) Five symbols per cell in the order centered Haar $\times$ p99.9 (the record, $Q \perp 1$), uncentered Haar $\times$ p99.9, Haar $\times$ supremum, Gaussian $\times$ p99.9, Gaussian $\times$ supremum, • = genuine under that construction; genuine counts 44/72, 49/72, 46/72, 52/72, 52/72. The percentage is the record excess divided by the mean null reading (ImageNet -33% to -5%; all cells -75% to +18%); the last column is the ImageNet excess of the supremum statistic under the same Haar null with its rank. The centered Haar construction reproduces the centered spectrum exactly, the uncentered one differs by a term of order $1/\sqrt{n}$ that moves the verdict only on the ten-class datasets; the Gaussian one does not at small n and inflates excesses for low-rank clouds (Table 2). The supremum is decided by a few extreme quadruples and disagrees with the 99.9th percentile in both directions on DINOv2-S/B/G ImageNet.

(b) Verdict under the five null $\times$ statistic constructions and the excess as a fraction of the null reading.

model	IN	C100	C10	DTD	FMNIST	MNIST	sup. IN exc. (r)
ViT-T	○○●● -5%	○○●● -5%	○○○○ -9%	●●●● -11%	○○○○ +18%	○○○○ -14%	−0.020 (199)
ViT-S	●●●● -16%	○○●● -0%	●●●● -33%	●●●● -16%	●●●● -21%	○○○○ -21%	−0.020 (200)
ViT-B	●●●● -23%	○○○○ +5%	●●●● -30%	●●●● -18%	●●●● -26%	○○○○ -5%	−0.022 (200)
ViT-L	●●●● -22%	●●●● -18%	●●●● -31%	●●●● -28%	●●●● -31%	○○●● -20%	−0.027 (200)
DINO-B	●●●● -27%	●●●● -28%	●●●● -32%	●●●● -22%	○○○○ -1%	○○○○ -21%	+0.000 (96)
DINOv2-S	●●●● -23%	●●●● -31%	●●●● -28%	●●●● -14%	○○○○ -19%	○○○○ +2%	+0.017 (0)
DINOv2-B	●●●● -33%	●●●● -38%	●●●● -59%	●●●● -31%	●●●● -40%	○○○○ -8%	+0.018 (0)
DINOv2-L	●●●● -31%	●●●● -48%	●●●● -75%	●●●● -37%	●●●● -19%	○○○○ -6%	+0.006 (0)
DINOv2-G	●●●● -20%	●●●● -56%	●●●● -75%	●●●● -34%	●●●● -26%	○○○○ -11%	+0.020 (0)
CLIP-B	○○●● -7%	●●●● -16%	○○●● -17%	●●●● -19%	○○○○ -19%	○○○○ +4%	−0.023 (199)
CLIP-L	●●●● -11%	●●●● -16%	○○●● -17%	●●●● -21%	○○○○ -22%	○○○○ -0%	−0.016 (198)
SigLIP-B	○○●● -9%	●●●● -18%	●●●● -23%	●●●● -19%	●●●● -23%	○○●● -16%	−0.015 (194)

Table 4: (continued)

(c) The same census on cosine geometry (L2-normalized centroids, geodesic distances, Haar null on the normalized cloud): robustness, not the record.

model	IN		C100		C10		DTD		FMNIST		MNIST	
	exc.	r	exc.	r	exc.	r	exc.	r	exc.	r	exc.	r
ViT-T	-0.012	200	-0.014 ^o	191	-0.043	198	-0.017	199	+0.007 ^o	73	-0.026 ^o	167
ViT-S	-0.017	200	-0.017	200	-0.073	200	-0.021	200	-0.049	200	-0.036 ^o	183
ViT-B	-0.022	200	-0.006 ^o	170	-0.066	200	-0.021	200	-0.041	199	-0.010 ^o	130
ViT-L	-0.021	200	-0.024	200	-0.080	200	-0.024	200	-0.066	200	-0.043	196
DINO-B	-0.021	200	-0.035	200	-0.070	200	-0.024	200	-0.006 ^o	116	-0.038 ^o	193
DINOv2-S	-0.014	200	-0.033	200	-0.084	200	-0.018	200	-0.054	200	-0.004 ^o	113
DINOv2-B	-0.013	200	-0.032	200	-0.140	200	-0.021	200	-0.081	200	-0.019 ^o	147
DINOv2-L	-0.010	200	-0.029	200	-0.151	200	-0.024	200	-0.051	200	-0.020 ^o	145
DINOv2-G	-0.006	200	-0.029	200	-0.169	200	-0.017	200	-0.064	200	-0.023 ^o	157
CLIP-B	-0.017	200	-0.023	200	-0.039	197	-0.031	200	-0.038 ^o	185	+0.002 ^o	93
CLIP-L	-0.017	200	-0.025	200	-0.045	199	-0.031	200	-0.038 ^o	192	-0.008 ^o	124
SigLIP-B	-0.013	200	-0.025	200	-0.053	200	-0.027	200	-0.052	199	-0.044	197

Table 4: (continued) (c) Cosine reading, BH over 72 cells: sign-negative 70/72, genuine 56/72 and 22/24 on ImageNet+CIFAR-100, verdict agreement with the Euclidean record 60/72 cells. Text under the same cosine protocol (padding-free embeddings): genuine 8/15: gpt2-m, gpt2-l, gpt2-xl, pythia-410m, pythia-1b, pythia-2b8, olmo-1b, gte-qwen2. The angular tree: sibling-triplet agreement on CIFAR-100 for DINOv2-L is 0.91 under cosine against 0.71 under Euclidean distance. (d) Two self-supervised ResNet-50 backbones read the same clustered structure as the ViTs.

(d) Backbones outside the ViT panel on the ImageNet centroid store (spectrum null, 20 replicates).

model	d	$\hat{\delta}$	excess (z)	reps above
Barlow Twins (ResNet-50)	2048	0.075	-0.039 (-4.8)	20/20
BYOL (ResNet-50)	2048	0.073	-0.020 (-2.4)	20/20

Table 4: (continued)

B.4 THE MODEL PANEL AND EXTRACTION

type	model	params
Supervised ViT (vision)	ViT-T (i21k)	5M
Supervised ViT (vision)	ViT-S (i21k)	22M
Supervised ViT (vision)	ViT-B (i21k)	86M
Supervised ViT (vision)	ViT-L (i21k)	307M
SSL (vision)	DINO-B	86M
SSL (vision)	DINOv2-S	22M
SSL (vision)	DINOv2-B	86M
SSL (vision)	DINOv2-L	307M
SSL (vision)	DINOv2-G	1.1B
Contrastive (vision)	CLIP-B	86M
Contrastive (vision)	CLIP-L	307M
Contrastive (vision)	SigLIP-B	86M
Causal LM	GPT-2 S	117M
Causal LM	GPT-2 M	345M
Causal LM	GPT-2 L	774M
Causal LM	GPT-2 XL	1.5B
Causal LM	Pythia-410M	410M
Causal LM	Pythia-1B	1.0B
Causal LM	Pythia-2.8B	2.8B
Causal LM	OLMo-1B	1.0B
Text embedder	BGE-base	110M
Text embedder	BGE-large	335M
Text embedder	GTE-base	110M
Text embedder	GTE-large	335M
Text embedder	GTE-Qwen2-1.5B	1.5B
Text embedder	E5-base	110M
Text embedder	E5-large	335M
Controls (vision)	DeiT-B/16 (IN-1k, no distillation)	86M
Controls (vision)	ViT-B/16 augreg (IN-1k)	86M
Controls (vision)	MERU ViT-S/B/L and CLIP twins	22M/86M/307M
Controls (vision)	Barlow Twins, BYOL (ResNet-50)	24M

Table 5: The model panel. 12 vision backbones (all in the census; the 10 with cached per-dataset features, all but DINO-B and SigLIP-B, form the task grid), 8 causal LMs, 7 text embedders, and the vision models used only as controls. Class centroids average 100 cached training images per class on ImageNet and the full cached split, 80–6,000 images per class, on the other datasets. Feature extraction: vision backbones use the encoder’s default pooled embedding (timm, num_classes=0, CLS or global pool per architecture); causal LMs use the last-token hidden state, extracted one prompt at a time (padding-free: left-padded batching alters GPT-2’s hidden states, Table 6); text embedders use their recommended pooling (CLS for BGE, mean for GTE/E5, last token for GTE-Qwen2); CLIP/SigLIP text towers use their text encoders.

B.5 IS TEXT THE SAME? THE TEXT CENSUS UNDER THE RECORD

Extraction, anisotropy and templates. Extraction is part of the measurement: left-padded batching moves GPT-2’s $\hat{\delta}$ by up to 0.018, of the order of the fragile excesses; the padding-free protocol is the record. Genuineness is not a proxy for anisotropy: GTE-base has the highest effective rank and is not genuine, and the only LLM-backed embedder tested (GTE-Qwen2-1.5B) is not genuine under the record. Causal-LM values are template-sensitive but the scale ordering is template-robust (Table 6).

(a) The census of record on the 1000 ImageNet class prompts: padding-free extraction, Haar null, 99.9th-percentile statistic, 200 replicates.

model	type	d	$\hat{\delta}_{99.9}$	excess	record exc./null	$r/200$	p	2×2	cosine exc.	r
GPT-2 S	causal LM	768	0.077	-0.013	-15%	197	.020	●●●○	+0.000°	101
GPT-2 M	causal LM	1024	0.052	-0.005°	-9%	170	.154	○○○○	-0.019	200
GPT-2 L	causal LM	1280	0.053	-0.023	-30%	200	.005	●●●●	-0.016	200
GPT-2 XL	causal LM	1600	0.048	-0.027	-36%	200	.005	●●●●	-0.018	200
Pythia-410M	causal LM	1024	0.067	-0.018	-21%	200	.005	●●●●	-0.024	200
Pythia-1B	causal LM	2048	0.066	-0.015	-18%	199	.010	●●●●	-0.026	200
Pythia-2.8B	causal LM	2560	0.066	-0.017	-20%	200	.005	●●●●	-0.023	200
OLMo-1B	causal LM	2048	0.051	-0.024	-32%	200	.005	●●●●	-0.032	200
BGE-base	embedder	768	0.078	+0.003°	+4%	26	.871	○○○○	-0.002°	151
BGE-large	embedder	1024	0.078	+0.002°	+3%	49	.756	○○○○	-0.003°	170
GTE-base	embedder	768	0.076	+0.004°	+6%	11	.945	○○○○	+0.001°	79
GTE-large	embedder	1024	0.075	+0.003°	+3%	36	.821	○○○○	-0.001°	131
GTE-Qwen2-1.5B	embedder	1536	0.078	+0.002°	+3%	58	.711	○○○○	-0.006	199
E5-base	embedder	768	0.074	+0.004°	+6%	15	.925	○○○○	+0.000°	97
E5-large	embedder	1024	0.071	+0.002°	+3%	44	.781	○○○○	-0.001°	119

Table 6: **In text, structure beyond the second moments depends on recipe and scale: genuine at every Pythia scale and at the larger GPT-2 sizes, at the margin for GPT-2 S, absent for GPT-2 M and for the sentence embedders on this probe.** (a) r = replicates above the real value, p the left-tail add-one p -value; genuine = BH-corrected $p \leq 0.05$ over the 15 models (°: not genuine); exc./null: excess as a fraction of the null reading; the 2×2 column gives the verdicts under Haar $\times$ p99.9, Haar $\times$ supremum, Gaussian $\times$ p99.9, Gaussian $\times$ supremum (● genuine); cosine: the same census on cosine geometry. Genuine 7/15; GPT-2 S is genuine at $p = .020$ with excess -0.013 and GPT-2 M is not ($p = .154$, -0.005), while L and XL are genuine under every construction. GTE-base has the highest effective rank and is not genuine, so genuineness is not a proxy for anisotropy; the only LLM-backed embedder (GTE-Qwen2-1.5B) is not genuine. (b) Causal-LM values are template-sensitive (range of raw $\hat{\delta}$ up to 0.065 for causal LMs against ≤ 0.012 for embedders; “gloss only” contains no class name) but the scale ordering is template-robust: with only 3 null replicates per cell the starred S and M cells are borderline, as the 200-replicate record in (a) shows, whereas the L/XL cells are not in doubt. (c) Precision is irrelevant (fp16 equals fp32 to four decimals); for GPT-2, left-padded batching changes the hidden states (mean cosine to the padding-free extraction 0.98) and moves $\hat{\delta}$ by up to 0.018, while OLMo handles left padding correctly and is invariant, which is why padding-free extraction is the record.

(b) Prompt templates: raw $\hat{\delta}$ per template and, for the four GPT-2 sizes, the template-conditioned excess over the spectrum null (3 replicates; *: $|z| \geq 2$) and its fraction of the null.

template	raw $\hat{\delta}$				GPT-2 excess (padding-free)				GPT-2 excess / null			
	BGE-b	E5-b	GPT-2 S	GPT-2 M	S	M	L	XL	S	M	L	XL
photo	0.124	0.114	0.131	0.091	-0.016*	-0.019*	-0.043*	-0.050*	-11%	-19%	-34%	-41%
name_only	0.118	0.119	0.145	0.122	-0.006	-0.007	-0.048*	-0.051*	-6%	-5%	-39%	-43%
image	0.125	0.117	0.110	0.110	-0.018*	-0.028*	-0.050*	-0.049*	-13%	-26%	-41%	-41%
closeup	0.126	0.120	0.086	0.100	+0.007	-0.018*	-0.044*	-0.049*	+5%	-18%	-36%	-40%
this_is	0.125	0.115	0.080	0.134	-0.032*	-0.022*	-0.042*	-0.046*	-19%	-22%	-34%	-38%
wild	0.115	0.116	0.089	0.104	-0.023*	-0.067*	-0.014	-0.010*	-16%	-43%	-11%	-8%
def_pair	0.119	0.108	0.105	0.081	-0.002	-0.019*	-0.036*	-0.035*	-1%	-21%	-28%	-29%
gloss_only	0.115	0.115	0.124	0.107	-0.020*	-0.030*	-0.033*	-0.036*	-14%	-31%	-27%	-29%
concept	0.119	0.120	0.112	0.112	-0.002	-0.003	-0.043*	-0.041*	-1%	-3%	-38%	-38%
discussion	0.121	0.116	0.105	0.098	-0.010	-0.013*	-0.041*	-0.040*	-7%	-17%	-35%	-35%
mean over templates					-0.012	-0.023	-0.040	-0.041				
templates at $ z \geq 2$					5 of 10	8 of 10	9 of 10	10 of 10				

Table 6: (continued)

(c) Extraction: the same 1000 prompts, only batch size and precision vary.

batch size	GPT-2 M		OLMo-1B	
	fp32 $\hat{\delta}$	fp16 $\hat{\delta}$	fp32 $\hat{\delta}$	fp16 $\hat{\delta}$
1 (no padding)	0.0816	0.0817	0.0817	0.0817
16	0.0996	0.0997	0.0817	0.0816
32	0.0905	0.0905	0.0817	0.0817

Table 6: (continued)

B.6 DOES THE PREMISE SURVIVE AT THE SAMPLE LEVEL? THE CENSUS ON PER-IMAGE FEATURES

(a) The census cells at the sample level.

model	CIFAR-100 (10 images/class, $n=1000$)					DTD (22 images/class, $n=1034$)				
	sup. $\hat{\delta}$	$\hat{\delta}_{99.9}$	excess	exc./null	$r/200$ (p)	sup. $\hat{\delta}$	$\hat{\delta}_{99.9}$	excess	exc./null	$r/200$ (p)
ViT-T	0.130	0.093	-0.0016°	-2%	134 (0.333)	0.124	0.087	-0.0004°	-0%	111 (0.448)
ViT-S	0.121	0.086	$+0.0051^\circ$	+6%	21 (0.896)	0.103	0.070	-0.0062	-8%	197 (0.020)
ViT-B	0.121	0.082	$+0.0031^\circ$	+4%	50 (0.751)	0.092	0.060	-0.0080	-12%	200 (0.005)
ViT-L	0.100	0.067	-0.0041°	-6%	185 (0.080)	0.092	0.056	-0.0073	-12%	200 (0.005)
DINO-B	0.104	0.070	-0.0042°	-6%	194 (0.035)	0.110	0.075	$+0.0008^\circ$	+1%	80 (0.602)
DINOv2-S	0.100	0.065	-0.0049	-7%	200 (0.005)	0.145	0.090	$+0.0058^\circ$	+7%	0 (1.000)
DINOv2-B	0.089	0.050	-0.0090	-15%	200 (0.005)	0.139	0.073	-0.0019°	-3%	162 (0.194)
DINOv2-L	0.087	0.043	-0.0131	-24%	200 (0.005)	0.131	0.064	-0.0060	-9%	200 (0.005)
DINOv2-G	0.102	0.036	-0.0232	-39%	200 (0.005)	0.136	0.065	-0.0079	-11%	200 (0.005)
CLIP-B	0.117	0.080	-0.0060°	-7%	184 (0.085)	0.125	0.089	$+0.0021^\circ$	+2%	71 (0.647)
CLIP-L	0.132	0.084	$+0.0037^\circ$	+5%	35 (0.826)	0.120	0.082	-0.0008°	-1%	124 (0.383)
SigLIP-B	0.113	0.078	-0.0074	-9%	199 (0.010)	0.114	0.079	-0.0044°	-5%	184 (0.085)

Table 7: **The sample-level reading, the object of prior latent-hyperbolicity claims, sits within null noise in most cells.** The instrument on ≈ 1000 stratified training images per cell (per-image features, not centroids): sup. $\hat{\delta}$ is the raw supremum δ_{norm} the literature reports; $\hat{\delta}_{99.9}$, excess, exc./null, $r/200$ and p follow Table 1 (Haar spectrum-matched null on the sample cloud, 99.9th-percentile statistic, 200 replicates); $^\circ$: not genuine under Benjamini–Hochberg over the 24 cells. The raw supremum lies in the band 0.087–0.145; the excess is not genuine in 14 of 24 cells and genuine in 10 (DINOv2-S, DINOv2-B, DINOv2-L, DINOv2-G, SigLIP-B on CIFAR-100; ViT-S, ViT-B, ViT-L, DINOv2-L, DINOv2-G on DTD), where it reaches at most -0.023 , that is -39% to -7% of the null reading.

(b) A published reading reproduced and calibrated: the setting of Khrulkov et al. (2020), Table 1, ResNet-34 row.

dataset	their δ_{rel}	our δ_{rel}	excess	rank r	largest p
CIFAR-10	0.26	0.275 ± 0.017	-0.0037 ± 0.0017	173/200	0.333
CIFAR-100	0.25	0.261 ± 0.013	-0.0087 ± 0.0025	198/200	0.030
CUB-200	0.25	0.270 ± 0.012	-0.0043 ± 0.0020	177/200	0.249
MiniImageNet	0.21	0.203 ± 0.013	-0.0234 ± 0.0018	200/200	0.005

(b) Penultimate features of an ImageNet-pretrained ResNet-34 (torchvision, no substitution) on class-balanced batches of 1500 points; their estimator (exact δ on the batch, $\delta_{\text{rel}} = 2\delta/\text{diam}$; mean $\pm$ s.d. over 10 batches) next to their published value, and on the same clouds the record instrument: excess over the centered Haar null (200 replicates, 99.9th-percentile statistic), the mean rank of the reading among the replicates and the largest left-tail p over the 10 batches. Every reproduced raw value is within 0.03 of the published one.

Table 7: (continued)

B.7 IS THE STRUCTURE DEEPER THAN A STAR? THE DEPTH TEST

Real-data readings by regime. ImageNet ($n=1000$, WordNet $K=30$) lies in the regime where false alarms are controlled: the real cloud is more hierarchical than its anisotropic matched star beyond the noise ($z \leq -2$) in 4 of 12 backbones (ViT-S, ViT-B, ViT-L, DINOv2-L), never less, under Gaussian and under Haar-resampled star hubs alike (Table 8, Figure 4a). CIFAR-100 ($n=100$, $K=20$) lies in the regime where false alarms are not controlled, where pure stars are declared more hierarchical in 17% of runs; its readings (3 of 12 with $z \leq -2$, none with $z \geq +2$) are reported without certification (Table 8). Validity domain of the test as run: $n \approx 1000$; at $n=100$ the false-alarm rate is 0% with $K=6$ (17 points per cluster) and 17% with $K \geq 12$, so the test is not validated below about ten points per cluster. The bar was fixed before the sweep; the restriction to the regime where false alarms are controlled was adopted after it.

A trained positive control. Table 8(d) gives the fine-tuned ViT-B/16 controls, cross-entropy alone and with a hierarchical term, and the decoupled comparison that separates them.

(a) The twelve backbones under the stars with Gaussian hubs: depth = excess B of the real centroids minus that of a matched star (negative = more hierarchical above the frame), with z against the combined spread; the WordNet cut at $K=10, 30$ and 60 superclasses and the balanced frame.

model	C100, $K=20$		ImageNet, anisotropic star				bal. frame
	iso. z	aniso. depth (z)	iso. $K=30$ z	$K=10$ z	$K=30$ depth (z)	$K=60$ z	real z
ViT-T	+0.02	-0.025 (-1.43)	-1.48	-1.70	-0.018 (-1.48)	-2.53	-2.08
ViT-S	+2.34	-0.008 (-0.51)	-1.95	-1.72	-0.028 (-2.36)	-1.78	-1.72
ViT-B	+4.47	+0.003 (+0.23)	-2.87	-2.57	-0.033 (-3.62)	-2.23	-2.82
ViT-L	+4.20	-0.017 (-1.49)	-3.17	-2.73	-0.033 (-4.04)	-4.69	-5.57
DINO-B	+0.86	-0.038 (-4.24)	-0.21	-0.62	-0.009 (-0.94)	-1.11	-0.98
DINOv2-S	+0.55	-0.020 (-1.61)	+0.94	+0.26	+0.008 (+0.66)	+0.61	+0.12
DINOv2-B	+5.40	+0.010 (+0.94)	-0.93	-0.46	-0.016 (-1.19)	-0.75	-0.32
DINOv2-L	+6.77	+0.013 (+1.20)	-2.27	-1.20	-0.028 (-2.39)	-1.25	-1.70
DINOv2-G	+3.76	-0.007 (-0.68)	-1.57	-1.00	-0.029 (-1.78)	-0.48	-0.93
CLIP-B	+1.75	-0.024 (-2.00)	-0.72	-1.55	-0.016 (-1.95)	-2.10	-1.29
CLIP-L	+5.68	-0.013 (-2.34)	+0.46	-1.26	-0.008 (-0.83)	-1.40	-0.68
SigLIP-B	+2.77	-0.021 (-2.17)	+0.05	-0.76	-0.011 (-0.90)	-1.66	-1.12

Table 8: **The depth test certifies in four ImageNet backbones under both matched stars that clusters are oriented toward their hubs, not that the hubs form a hierarchy, and no hub hierarchy is found in the nine backbones where the decoupled control detects an implanted one, and the test is blind in the other three: none fires once cluster orientations are randomized, whereas a three-level hierarchy implanted at the same noise level is detected and survives that randomization (Table 9); imposing the geometry does not create the structure, and leaf-label supervision can produce it or not.** (a) Isotropic star: Gaussian clouds with each superclass’s RMS spread (the earlier, withdrawn construction); anisotropic star: within each superclass a Haar sample with the cloud’s own covariance, hubs Gaussian at the real hub radius; Haar-resampled hubs: the same clusters with the hubs drawn as a Haar resample of the real hubs. Hub-randomizing Haar null, 10 star seeds; frames: CIFAR-100 coarse labels, ImageNet WordNet cut. The test controls false alarms at $n=1000$ only (Table 9); CIFAR-100 readings ($n=100$) are reported without certification. Certified on ImageNet ($n=1000$, the regime where false alarms are controlled): 4 of 12 backbones (ViT-S, ViT-B, ViT-L, DINOv2-L) with z from -2.36 to -4.04 ; the withdrawn isotropic star reached $z=+6.77$ on CIFAR-100. With the hubs of the matched star drawn as a Haar resample of the real hubs (exact hub spectrum, 10 star seeds) the certified set is ViT-S, ViT-B, ViT-L, DINOv2-L, with z from -2.04 to -3.73 ; r_{star} counts the star seeds whose excess B lies at or below the real one, a rank whose resolution is one part in eleven and which does not separate the certified backbones from the others. Balanced frame (complete-linkage WordNet cut with the smallest variance of cluster sizes, chosen before any depth run): certified ViT-T, ViT-B, ViT-L. K sweep: CIFAR-100 $K=5$: iso 0/12 at $z \leq -2$, 7/12 at $z \geq +2$; aniso 0/12 at $z \leq -2$, 0/12 at $z \geq +2$. CIFAR-100 $K=10$: iso 0/12 at $z \leq -2$, 9/12 at $z \geq +2$; aniso 1/12 at $z \leq -2$, 0/12 at $z \geq +2$. CIFAR-100 $K=20$: iso 0/12 at $z \leq -2$, 8/12 at $z \geq +2$; aniso 3/12 at $z \leq -2$, 0/12 at $z \geq +2$. ImageNet $K=10$: iso 0/12 at $z \leq -2$, 0/12 at $z \geq +2$; aniso 2/12 at $z \leq -2$, 0/12 at $z \geq +2$. ImageNet $K=30$: iso 3/12 at $z \leq -2$, 0/12 at $z \geq +2$; aniso 4/12 at $z \leq -2$, 0/12 at $z \geq +2$. ImageNet $K=60$: iso 3/12 at $z \leq -2$, 0/12 at $z \geq +2$; aniso 4/12 at $z \leq -2$, 0/12 at $z \geq +2$. (b) Supervision on leaf labels can produce the depth (the augreg recipe is certified under both stars) and need not (DeiT-B is not detected): the hierarchical label set of IN-21k is not what makes the census ViTs deep; their WordNet alignment is in Table 11.

(a') The same backbones under the star whose hubs are a Haar resample of the real hubs (ImageNet, $K=30$); last column: the decoupling control, the real hubs kept and each cluster's offsets rotated by an independent Haar rotation (mean $z \pm$ s.d. over 10 seeds; seeds certified at $z \leq -2$).

model	star B	depth (z)	r_{star}	decoupled z (cert.)
ViT-T	+0.0011	-0.0184 (-1.45)	0/10	-1.69 ± 0.28 (1/10)
ViT-S	+0.0010	-0.0257 (-2.04)	0/10	-1.30 ± 0.19 (0/10)
ViT-B	+0.0012	-0.0347 (-3.01)	0/10	-1.40 ± 0.25 (0/10)
ViT-L	+0.0000	-0.0337 (-3.73)	0/10	-1.61 ± 0.19 (0/10)
DINO-B	+0.0002	-0.0092 (-1.02)	0/10	$+1.75 \pm 0.16$ (0/10)
DINOv2-S	+0.0007	+0.0083 (+0.71)	10/10	$+1.38 \pm 0.14$ (0/10)
DINOv2-B	+0.0018	-0.0155 (-1.18)	0/10	$+0.41 \pm 0.11$ (0/10)
DINOv2-L	-0.0007	-0.0267 (-2.21)	0/10	-0.20 ± 0.14 (0/10)
DINOv2-G	-0.0021	-0.0263 (-1.50)	0/10	-0.04 ± 0.07 (0/10)
CLIP-B	+0.0012	-0.0171 (-1.48)	1/10	-1.25 ± 0.09 (0/10)
CLIP-L	+0.0018	-0.0090 (-0.74)	2/10	-0.88 ± 0.22 (0/10)
SigLIP-B	+0.0024	-0.0121 (-0.78)	1/10	-0.84 ± 0.18 (0/10)

Table 8: (continued)

(b) Two ViT-B/16 supervised on ImageNet-1k leaf labels only, through the census protocol, against the IN-21k census backbone.

model	IN excess (r)	C100 excess (r)	depth, Gaussian hubs (z)	depth, Haar hubs (z)
DeiT-B, IN-1k	-0.0073 (200)	-0.0257 (200)	-0.0213 (-1.26)	-0.0197 (-1.21)
ViT-B, augreg, IN-1k	-0.0070 (199)	-0.0117 (187)	-0.0249 (-3.03)	-0.0290 (-2.65)
ViT-B, augreg, IN-21k (census)	-0.0162 (200)	+0.0057 (43)	-0.0335 (-3.62)	-0.0347 (-3.01)

Table 8: (continued)

(c) A hyperbolic backbone and its Euclidean twin under the census of record and the depth test.

		CIFAR-100 ($K=20$, $n=100$: false alarms uncontrolled)			ImageNet ($K=30$, $n=1000$: false alarms controlled)		
model	input	exc. (r)	$\hat{\delta}_{99.9}$ nat./Eucl.	depth z	exc. (r)	$\hat{\delta}_{99.9}$ nat./Eucl.	depth z
MERU ViT-S	images	−0.020 (199)	0.117/0.117	−1.52	−0.008° (186)	0.088/0.088	−1.54
CLIP ViT-S	images	−0.028 (200)	0.106/0.107	−3.95	−0.006° (171)	0.110/0.088	−1.85
MERU ViT-B	images	−0.020 (200)	0.114/0.114	−2.16	−0.010 (196)	0.087/0.087	−0.86
CLIP ViT-B	images	−0.030 (200)	0.105/0.103	−3.28	−0.009° (187)	0.107/0.085	−0.99
MERU ViT-L	images	−0.012° (195)	0.118/0.118	−0.29	−0.007° (184)	0.088/0.088	−0.52
CLIP ViT-L	images	−0.016 (199)	0.123/0.112	−1.68	−0.010 (197)	0.104/0.084	−1.21
MERU ViT-S	prompts	−0.022 (200)	0.086/0.086	−0.67	−0.003° (155)	0.085/0.086	+0.02
CLIP ViT-S	prompts	−0.023 (200)	0.085/0.082	−0.78	+0.006° (15)	0.093/0.087	+0.82
MERU ViT-B	prompts	−0.012 (200)	0.096/0.096	−0.70	−0.001° (132)	0.089/0.090	+0.76
CLIP ViT-B	prompts	−0.017 (200)	0.094/0.088	+0.31	+0.008° (2)	0.100/0.092	+0.18
MERU ViT-L	prompts	−0.006° (190)	0.102/0.102	+0.58	+0.001° (75)	0.093/0.093	+0.52
CLIP ViT-L	prompts	−0.012 (200)	0.098/0.092	+0.48	+0.006° (5)	0.101/0.092	+0.39

Table 8: (continued) (c) MERU (Desai et al., 2023) embeds images and text on the Lorentz hyperboloid with an entailment objective; its released twin is a CLIP baseline with the same ViT, RedCaps data and recipe minus the hyperbolic lift and entailment loss. Both pass through the instrument unchanged: class centroids of the projected embeddings (MERU: space-like hyperboloid coordinates; CLIP: unit vectors), Euclidean distances; exc. (r): excess over the Haar spectrum-matched null with the number of 200 replicates above the real value (°: not genuine under BH over the 24 cells); nat./Eucl.: the same statistic in the model’s own metric (Lorentz distance between tangent-space-mean centroids for MERU, angular for CLIP) against the Euclidean one; depth z : the anisotropic matched-star test of panel (a) (10 star seeds), validated at $n=1000$ only. On ImageNet images the excess runs from −0.007 to −0.010 for MERU and from −0.006 to −0.010 for its twin across ViT-S/B/L; MERU’s Lorentz-metric $\hat{\delta}_{99.9}$ differs from its Euclidean value by at most 0.0002; depth z runs from −1.54 to −0.52 for MERU and from −1.85 to −0.99 for CLIP, so neither is more hierarchical than a matched star and MERU is never ahead of its twin. Its embeddings sit in the near-flat regime of the hyperboloid: learned curvature $c=0.10$, spatial norm times $\sqrt{c}$ at the median 0.258–0.279 (95th percentile 0.263–0.287), and Lorentz-to-Euclidean pairwise distance ratio 0.9972–0.9991 at the median (0.9988–0.9997 on the class centroids), so the control tests the training objective, not a curved geometry. MERU’s hierarchy is a generic→specific (text→image) partial order, not a class taxonomy. (d) Five epochs, AdamW, lr 10^{-5} encoder and 10^{-3} heads, batch 256, seed 0; centroids of the census subset; decoupling over 10 seeds (fraction certified in parentheses). The hierarchical model is certified on both frames and keeps firing once decoupled; the leaf-CE model and the frozen checkpoint are certified intact but do not fire once decoupled on the frame of record, and the frozen checkpoint fires in 7 of 10 decoupled seeds on the balanced frame. The pre-set criterion (leaf-CE and frozen not certified intact) is not met, because ViT-B is certified by alignment; the discriminating comparison is the decoupled one.

(d) A trained positive control: the census ViT-B/16 fine-tuned on the full ImageNet-1k training set with leaf cross-entropy alone and with an added hierarchical cross-entropy at the WordNet 30/6/2 cuts (weights 1/2/4), one seed, identical batches; census excess under the centered Haar null, depth test and decoupling control on both frames.

ViT-B/16	excess (r)	z , WordNet-30	dec. z (cert.)	z , balanced	dec. z (cert.)
frozen (census)	−0.0162 (200)	−3.62	−1.40 (0.0)	−2.82	−2.10 (0.7)
leaf cross-entropy	−0.0243 (200)	−2.45	−0.72 (0.0)	−2.09	−1.38 (0.0)
leaf + hierarchical cross-entropy	−0.0293 (200)	−2.97	−2.38 (1.0)	−3.57	−3.51 (1.0)

Table 8: (continued)

(e) Radial control: the depth test on the same clouds after L2-normalizing every centroid and after removing the radial component of every offset (its projection onto the hub direction), under both matched stars, with the decoupling control on each transformed cloud.

model	L2-normalized			radial component removed			
	z aniso	z Haar-hub	decoupled z (cert.)	z aniso	z Haar-hub	decoupled z (cert.)	
ViT-T	+0.27	-0.17	-0.78 (0.0)	-2.06	-2.07	-1.74 (0.3)	
ViT-S	-0.76	-0.93	-1.16 (0.0)	-2.26	-2.08	-1.25 (0.0)	
ViT-B	-2.45	-2.52	-0.57 (0.0)	-3.60	-3.57	-1.40 (0.0)	
ViT-L	-2.00	-2.23	-1.41 (0.1)	-3.99	-3.47	-1.47 (0.0)	
DINO-B	+0.82	+0.95	+2.00 (0.0)	-1.61	-1.89	+1.91 (0.0)	
DINOv2-S	+1.27	+1.38	+1.28 (0.0)	+0.61	+0.58	+1.43 (0.0)	
DINOv2-B	-0.07	-0.01	+1.28 (0.0)	-1.18	-1.19	+0.46 (0.0)	
DINOv2-L	-0.46	-0.50	+0.89 (0.0)	-2.59	-2.28	-0.01 (0.0)	
DINOv2-G	-0.78	-0.76	+0.88 (0.0)	-1.75	-1.66	+0.11 (0.0)	
CLIP-B	-1.42	-0.93	-1.38 (0.0)	-1.53	-1.06	-1.19 (0.0)	
CLIP-L	-1.57	-1.12	-0.82 (0.0)	-1.05	-0.89	-0.92 (0.0)	
SigLIP-B	-1.13	-0.88	-0.51 (0.0)	-1.12	-1.07	-0.89 (0.0)	

Table 8: (continued)

B.8 HOW MUCH DEPTH CAN THE TEST SEE? POWER ON SYNTHETIC AND IMPLANTED HIERARCHIES



Figure 6: **A two-level implant on the real ImageNet clouds is missed at the real within-cluster spread and detected once the spread is shrunk.** Detection rate at $z \leq -2$ against implant strength s with the real within-cluster spread (solid) and shrunk (dashed): at full strength 0.05 against 1.00, and 0.00 at $s = 0$ with the real spread.

(a) Synthetic hierarchies at $d=768$: two- and three-level hierarchies (pooled) and pure stars, within/between noise ratios 0.1/0.3/0.6, isotropic and anisotropic clusters pooled, 5 seeds.

frame	n	K	power @ ratio 0.1	0.3	0.6	star $z \leq -2$	star $z \geq +2$
top-level, isotropic star	100	6	0.00	0.00	0.00	0.00	0.17
top-level, isotropic star	100	12	0.00	0.00	0.00	0.00	0.13
top-level, isotropic star	100	20	0.00	0.00	0.00	0.00	0.17
top-level, isotropic star	100	30	0.00	0.00	0.00	0.00	0.17
top-level, isotropic star	1000	6	0.00	0.00	0.00	0.00	0.17
top-level, isotropic star	1000	12	0.00	0.00	0.00	0.00	0.17
top-level, isotropic star	1000	20	0.00	0.00	0.00	0.00	0.17
top-level, isotropic star	1000	30	0.00	0.00	0.00	0.00	0.17
leaf clusters, anisotropic star	100	6	1.00	1.00	1.00	0.00	0.00
leaf clusters, anisotropic star	100	12	1.00	1.00	1.00	0.17	0.00
leaf clusters, anisotropic star	100	20	1.00	1.00	1.00	0.17	0.00
leaf clusters, anisotropic star	1000	6	1.00	1.00	1.00	0.00	0.00
leaf clusters, anisotropic star	1000	12	1.00	1.00	1.00	0.00	0.00
leaf clusters, anisotropic star	1000	20	1.00	1.00	0.90	0.00	0.00
leaf clusters, anisotropic star	1000	30	1.00	1.00	0.70	0.00	0.00

Table 9: **The depth test has full power on synthetic hierarchies at the leaf frame and none at the top-level frame used on real backbones.** (a) Power = fraction of hierarchy runs with $z \leq -2$; star columns = false-alarm rates in each direction. Top-level frame: the test receives the K super-cluster labels, as the earlier isotropic test did on the real backbones; its hub null keeps every frame cluster intact and only rearranges the K hubs, so hierarchy below the frame is invisible by construction. Leaf frame: the test receives the finest cluster labels with the anisotropic matched star (configurations with more leaves than points are infeasible and omitted). Pre-set bar: power ≥ 0.8 for two- and three-level hierarchies at noise ratios ≤ 0.3 for both n , and false alarms $\leq 5\%$ in each direction. Leaf frame: met at $n=1000$ (power 1.00 at ratios ≤ 0.3 and 0.90 at 0.6; false alarms 0% and 0%), failed at $n=100$, where $K \geq 12$ (under ten points per cluster) gives 17% false alarms at $z \leq -2$. Top-level frame: power 0.00, false alarms 16% pooled. (b) Six super-hubs of five, hub = s -super-hub + $(1-s+0.4s)$ -own Gaussian draw, hub cloud rescaled to the real RMS radius; the depth test of Table 8 runs unchanged, 5 implant seeds per s . hits: implant seeds with $z \leq -2$ (of 5); s^* : first s with ≥ 4 of 5; within/between: RMS of the within-cluster offsets over the point-weighted RMS of the hub displacements (1.3 to 3.9 on the real clouds, against 0.1–0.6 in the synthetic sweep); tight: the same implant after shrinking the offsets to within/between = 0.6 (2 seeds). Power (fraction of runs with $z \leq -2$) by s : $s=0$: 0.00, $s=0.25$: 0.00, $s=0.5$: 0.00, $s=0.75$: 0.00, $s=1$: 0.05; false alarms at $s=0$: 0 of 60. Under the Haar-hub star the same implants give 0 of 60 false alarms at $s=0$ and 1 of 60 detections at $s=1$ (power 0.02). Balanced frame: the $K=30$ agglomerative cut of the WordNet distance matrix (complete linkage) with the smallest variance of cluster sizes, chosen and logged before any depth run; sizes 1–154 classes against 1–316 for the frame of record. (c) Implanted alignment is detected in 36 of 60 runs at full strength (60%) and in 1 of 60 at zero; the pre-set bar of 0.8 power at full strength is not met, and the power is backbone-dependent: every seed for the supervised ViTs and CLIP-B, none for DINOv2-S/B/L.

(b) Implanted two-level trees on the real ImageNet centroids: the hub arrangement replaced at strength s , every within-cluster offset kept; hits = implant seeds with $z \leq -2$ at $s=0/0.5/1$, $\bar{z}_1$ = mean z at $s=1$, w/b = within/between spread.

model	WordNet-30 frame of record, Gaussian-hub star						Haar-hub star		balanced frame (pre-specified)		
	real z	w/b	hits	s^*	$\bar{z}_1$	tight z at $s=0/0.5/1$	hits $s=0/1$	$\bar{z}_1$	real z	hits	s^*
ViT-T	-1.48	1.5	0/0/0	-	+0.12	-0.60/-1.85/-9.22	0/0	+0.04	-2.08	0/0/1	-
ViT-S	-2.36	1.9	0/0/0	-	-1.22	-0.23/-3.45/-10.05	0/0	-0.70	-1.72	0/0/0	-
ViT-B	-3.62	2.0	0/0/0	-	+0.37	-1.03/-3.92/-11.45	0/0	+0.41	-2.82	0/0/0	-
ViT-L	-4.04	1.9	0/0/0	-	-1.37	-1.61/-4.37/-12.68	0/0	-0.98	-5.57	0/0/0	-
DINO-B	-0.94	2.1	0/0/0	-	-0.31	-1.61/-5.14/-12.09	0/0	-0.29	-0.98	0/0/0	-
DINOv2-S	+0.66	3.0	0/0/0	-	-0.79	-0.19/-3.06/-11.48	0/0	-0.79	+0.12	0/0/0	-
DINOv2-B	-1.19	3.7	0/0/0	-	+0.41	-0.01/-3.07/-11.17	0/0	+0.55	-0.32	0/0/0	-
DINOv2-L	-2.39	3.9	0/0/0	-	+0.20	+0.21/-2.67/-11.40	0/0	+0.21	-1.70	0/0/0	-
DINOv2-G	-1.78	3.9	0/0/0	-	+1.30	+0.24/-3.89/-11.58	0/0	+1.23	-0.93	0/0/0	-
CLIP-B	-1.95	1.3	0/0/0	-	-1.21	-1.67/-3.81/-11.38	0/0	-0.72	-1.29	0/0/0	-
CLIP-L	-0.83	1.5	0/0/3	-	-2.18	-2.43/-5.76/-13.51	0/1	-1.72	-0.68	0/0/0	-
SigLIP-B	-0.90	1.5	0/0/0	-	-1.55	-2.09/-5.04/-11.03	0/0	-1.07	-1.12	0/0/0	-

Table 9: (continued)

(c) Implanted hub alignment on the real ImageNet clouds: from the decoupled cloud of the decoupling control, each cluster's principal axis rotated toward its hub direction by a fraction s of the angle (hubs and within-cluster spectra unchanged); detection rate at $z \leq -2$ over 5 seeds per backbone and pooled.

model	$s=0$	$s=0.25$	$s=0.5$	$s=0.75$	$s=1$
ViT-T	0.2	0.8	0.8	1.0	1.0
ViT-S	0.0	0.4	1.0	1.0	1.0
ViT-B	0.0	0.8	1.0	1.0	1.0
ViT-L	0.0	0.6	1.0	1.0	1.0
DINO-B	0.0	0.0	0.0	0.0	0.8
DINOv2-S	0.0	0.0	0.0	0.0	0.0
DINOv2-B	0.0	0.0	0.0	0.0	0.0
DINOv2-L	0.0	0.0	0.0	0.2	0.0
DINOv2-G	0.0	0.0	0.0	0.2	0.4
CLIP-B	0.0	0.0	1.0	1.0	1.0
CLIP-L	0.0	0.2	0.2	0.4	0.4
SigLIP-B	0.0	0.0	1.0	0.8	0.6
pooled (12 backbones $\times$ 5 seeds)	0.02	0.23	0.50	0.55	0.60

Table 9: (continued)

(d) A deep hierarchy at the real noise level: synthetic clouds with ViT-L's real ImageNet spectrum and a three-level implanted hierarchy (nested 2/6/30 cuts of the frame) at ViT-L's real within/between ratio, a flat two-level control (30 iid hubs, same spectrum and ratio), 5 seeds each, and the WordNet Poincaré embeddings of Nickel and Kiela trained on the transitive closure of the tree over the 1000 ImageNet leaves, read with Euclidean distances on the ball coordinates.

cloud	seed	d	excess (r)	depth z	decoupled z (cert.)
three-level hierarchy, ViT-L spectrum	0	1024	-0.0081 (199)	-1.80	-4.82 $\pm$ 0.73 (1.0)
three-level hierarchy, ViT-L spectrum	1	1024	-0.0095 (199)	-2.45	-4.44 $\pm$ 0.58 (1.0)
three-level hierarchy, ViT-L spectrum	2	1024	-0.0090 (199)	-2.38	-4.19 $\pm$ 0.60 (1.0)
three-level hierarchy, ViT-L spectrum	3	1024	-0.0099 (199)	-2.08	-4.48 $\pm$ 0.67 (1.0)
three-level hierarchy, ViT-L spectrum	4	1024	-0.0090 (199)	-2.20	-4.28 $\pm$ 0.54 (1.0)
flat control, ViT-L spectrum	0	1024	-0.0198 (200)	-0.08	-1.61 $\pm$ 0.27 (0.1)
flat control, ViT-L spectrum	1	1024	-0.0202 (200)	-0.32	-1.61 $\pm$ 0.31 (0.1)
flat control, ViT-L spectrum	2	1024	-0.0196 (200)	-0.20	-1.76 $\pm$ 0.37 (0.2)
flat control, ViT-L spectrum	3	1024	-0.0197 (200)	-0.69	-1.74 $\pm$ 0.33 (0.2)
flat control, ViT-L spectrum	4	1024	-0.0192 (200)	-0.27	-1.68 $\pm$ 0.39 (0.2)
WordNet Poincaré, $d=10$	0	10	+0.0830 (0)	+2.36	+0.32 $\pm$ 0.58 (0.0)
WordNet Poincaré, $d=50$	0	50	+0.0354 (0)	+1.53	-2.22 $\pm$ 0.65 (0.6)

Table 9: (continued)

(e) Power per backbone for a deep hierarchy at the backbone's own noise level: the three-level synthetic hierarchy of (d) built with each backbone's real ImageNet spectrum and within/between ratio; depth test at $K=30$ (10 star seeds) on the intact cloud and the decoupling control (10 seeds) on each.

model	within/between	seeds	power	mean z	decoupled power	decoupled mean z
ViT-T	1.55	5	0.00	-0.48	0.60	-2.17
ViT-S	1.93	5	0.00	-1.04	0.90	-2.77
ViT-B	1.96	5	0.00	-1.74	1.00	-4.21
ViT-L	1.88	5	0.80	-2.18	1.00	-4.44
DINO-B	2.12	5	0.00	-1.21	0.28	-1.64
DINOv2-S	3.00	20	0.00	-0.98	0.01	-0.90
DINOv2-B	3.71	20	0.45	-1.93	0.83	-2.44
DINOv2-L	3.88	20	0.95	-2.66	1.00	-3.19
DINOv2-G	3.92	20	1.00	-2.88	1.00	-3.54
CLIP-B	1.33	5	0.20	-1.69	1.00	-5.00
CLIP-L	1.49	5	1.00	-2.83	1.00	-4.98
SigLIP-B	1.54	5	1.00	-2.49	1.00	-4.86

Table 9: (continued)

(f) False alarms of the balanced frame: the flat control of (d), ViT-L’s spectrum and ratio, 5 seeds, read on the balanced thirty-superclass frame intact and decoupled (10 seeds each), with the flat hubs assigned by the balanced frame itself or by the WordNet-30 frame of record.

flat control	intact fired	mean z	decoupled fired	mean z	range
matched hubs (balanced frame)	0/5	−0.46	0/50	−0.65	+0.26 to −1.52
mismatched hubs (WordNet-30 frame)	5/5	−3.13	38/50	−2.33	−1.56 to −3.75

Table 9: (continued)

B.9 IS THE ISLAND REAL? THE TREE MAP UNDER SIX CONFIGURATIONS

Threshold sensitivity of the selection criterion of §5.4: the selected configuration is stable for degeneracy bounds in $[0.45, 0.50]$ on ImageNet and $[0.40, 0.50]$ on CIFAR-100; tighter bounds select cosine-complete on ImageNet (gap 0.13) and cosine-Ward on CIFAR-100 (gap 0.56), and in every case the label-free gap is moderate and configuration-sensitive, never a categorical island. The criterion must be applied per dataset: the configuration it selects on DBpedia (Euclidean, average linkage, where nothing is degenerate) is exactly the degenerate naive configuration on ImageNet, so a criterion transferred across datasets would reintroduce the island.

(a) Selection diagnostics and mean pairwise ARI at the reference cut.

dataset	configuration	max frac.	CPCC	mean pairwise ARI at the reference cut		
				B/L/G vs block	S, DINO-B vs block	within
ImageNet	euclid-average (deg.)	0.97	0.761	0.03	0.08	0.64
ImageNet	euclid-complete (deg.)	0.83	0.606	0.03	0.17	0.44
ImageNet	euclid-ward	0.48	0.464	0.15	0.38	0.53
ImageNet	cosine-average (selected)	0.40	0.691	0.38	0.35	0.48
ImageNet	cosine-complete	0.32	0.539	0.13	0.17	0.50
ImageNet	cosine-ward	0.37	0.464	0.17	0.39	0.55
CIFAR-100	euclid-average (deg.)	0.80	0.798	0.00	0.41	0.57
CIFAR-100	euclid-complete (deg.)	0.66	0.722	0.17	0.40	0.54
CIFAR-100	euclid-ward	0.33	0.534	0.37	0.65	0.63
CIFAR-100	cosine-average (deg.)	0.53	0.777	0.30	0.54	0.58
CIFAR-100	cosine-complete (selected)	0.39	0.698	0.39	0.51	0.63
CIFAR-100	cosine-ward	0.14	0.584	0.56	0.62	0.61

Table 10: **The categorical island is a chaining artifact of the naive cut; a moderate gap remains under every admissible configuration and closes only in triplet agreement under cosine-average.** (a) Six configurations (metric $\times$ linkage) per dataset with the two quantities that drive the selection criterion of §5.4: the degeneracy diagnostic (largest-cluster fraction at the reference cut, worst model) and the mean cophenetic fidelity (CPCC) of each model’s dendrogram to its own distance matrix; configurations with max-cluster fraction > 0.5 are excluded and, among the rest, the highest CPCC selects the marked row; then the mean pairwise ARI between the cuts at 30 (ImageNet) or 20 (CIFAR-100) clusters. (b) Dendrograms rebuilt from the census cache: ARI at the cut, Pearson correlation between the two trees’ cophenetic distance vectors, and agreement on which pair merges first over 10^4 random class triplets. Under the naive ImageNet configuration, cross-family ARI between the supervised ViTs and the contrastive VLMs reaches 0.82 while every DINOv2-vs-block pair stays at or below 0.06, and the worst model’s largest cluster holds 97% of the classes. Under the selected configuration the DINOv2 family agrees with the block at 0.38 against 0.48 within the block. Under the other 3 admissible ImageNet configurations the DINOv2-vs-block agreement is 0.13–0.17 against 0.50–0.55 within the block; over the admissible configurations and the three measures the median fraction of within-block agreement missing is 0.34, about a third.

(b) Three measures of DINOv2-B/L/G against the supervised+contrastive block, each written as vs block / within block.

dataset	configuration	ARI at the cut	cophenetic corr.	triplet agreement
ImageNet	euclid-average (deg.)	0.03 / 0.64	0.36 / 0.80	0.47 / 0.74
ImageNet	euclid-complete (deg.)	0.03 / 0.44	0.20 / 0.66	0.41 / 0.65
ImageNet	euclid-ward	0.16 / 0.53	0.62 / 0.84	0.67 / 0.85
ImageNet	cosine-average (selected)	0.39 / 0.48	0.48 / 0.78	0.77 / 0.76
ImageNet	cosine-complete	0.13 / 0.49	0.18 / 0.72	0.36 / 0.71
ImageNet	cosine-ward	0.17 / 0.55	0.59 / 0.85	0.69 / 0.86
CIFAR-100	euclid-average (deg.)	0.00 / 0.57	0.15 / 0.78	0.41 / 0.74
CIFAR-100	euclid-complete (deg.)	0.17 / 0.54	0.23 / 0.72	0.41 / 0.66
CIFAR-100	euclid-ward	0.37 / 0.63	0.63 / 0.77	0.70 / 0.79
CIFAR-100	cosine-average (deg.)	0.30 / 0.58	0.42 / 0.72	0.51 / 0.64
CIFAR-100	cosine-complete (selected)	0.39 / 0.63	0.46 / 0.64	0.50 / 0.61
CIFAR-100	cosine-ward	0.56 / 0.61	0.74 / 0.73	0.76 / 0.76

Table 10: (continued)

B.10 WHOSE TAXONOMY? WORDNET ALIGNMENT, SUPERCLASS RECOVERY, DBPEDIA AND HIERARCAPS

On HierarCaps (Alper & Averbuch-Elor, 2024) (1000 four-level caption chains), text encoders show the same reading with no class set: the projection radius orders levels within a chain, sibling triplets are resolved at 0.83–0.97 (chance 0.5), and the leaf excess repeats the text census (Table 11).

(a) Alignment with WordNet on ImageNet.

model	Spearman ρ , inter-centroid vs WordNet distance				raw $\hat{\delta}$ of WordNet / random groups
	ρ_{WN}	shuffle	Gaussian	single image	
ViT-T	+0.52	+0.02	+0.01	–	0.146 / 0.159
ViT-S	+0.50	+0.00	–0.00	–	0.110 / 0.130
ViT-B	+0.49	+0.03	+0.01	–	0.092 / 0.120
ViT-L	+0.53	+0.01	+0.03	+0.25	0.104 / 0.117
DINO-B	+0.36	–0.01	–0.01	–	0.084 / 0.117
DINOv2-S	+0.22	+0.01	+0.01	–	0.077 / 0.100
DINOv2-B	+0.18	–0.01	+0.01	–	0.068 / 0.069
DINOv2-L	+0.18	–0.00	–0.03	+0.08	0.060 / 0.058
DINOv2-G	+0.20	–0.03	–0.01	+0.03	0.054 / 0.056
CLIP-B	+0.57	+0.00	+0.01	–	0.126 / 0.174
CLIP-L	+0.59	+0.01	+0.03	+0.26	0.117 / 0.162
SigLIP-B	+0.58	–0.01	+0.01	–	0.118 / 0.158
DeiT-B (IN-1k)	+0.08	+0.01	–	–	–
ViT-B (augreg, IN-1k)	+0.52	+0.01	–	–	–

Table 11: **Alignment with the human taxonomy follows supervision and recipe, the superclasses are recovered by every family, and the recovery is not WordNet circularity.** (a) ρ_{WN} : Spearman correlation between inter-centroid and WordNet tree distances on ImageNet, with the label-shuffle and Gaussian-cloud controls; single image: the same correlation on single-image distances (one image per class); WordNet / random groups: raw $\hat{\delta}$ of WordNet-coherent against random class groupings. (b) ARI of the 20-cluster cut against the true CIFAR-100 superclasses per configuration (degenerate configurations marked; the criterion-selected one is cosine-complete). Paired within each of the 3 admissible configurations, the label-supervised ViT-B recovers more than each DINOv2 model in 11 of 12 comparisons; DINOv2 itself recovers the superclasses at ARI 0.19–0.52, and the best cut over models and configurations reaches 0.61 against zero for permuted labels. Circularity controls: WordNet-coherent groupings read a lower raw δ than random ones in 11 of 12 models, and single-image distances still correlate with WordNet at up to +0.26. The two leaf-label ViT-B/16 of Table 8: the augreg recipe is aligned with WordNet like the IN-21k ViTs (ρ_{WN} +0.52) while DeiT-B is nearly unaligned (+0.08) yet recovers the CIFAR-100 superclasses at ARI up to 0.56, so WordNet alignment is recipe-dependent even among leaf-supervised ViTs.

(b) Recovery of the CIFAR-100 superclasses: ARI of the 20-cluster cut per configuration († : degenerate on CIFAR-100, euclid-average being the naive configuration; * : criterion-selected).

model	euclid-avg †	euclid-ward	cosine-avg †	cosine-complete *	cosine-ward
ViT-T	0.31	0.38	0.29	0.33	0.33
ViT-S	0.35	0.56	0.37	0.47	0.56
ViT-B	0.39	0.61	0.43	0.45	0.57
ViT-L	0.38	0.55	0.49	0.48	0.60
DINO-B	0.22	0.48	0.28	0.38	0.54
DINOv2-S	0.12	0.51	0.32	0.33	0.49
DINOv2-B	0.02	0.47	0.35	0.42	0.51
DINOv2-L	0.01	0.32	0.25	0.50	0.48
DINOv2-G	0.00	0.19	0.08	0.20	0.52
CLIP-B	0.18	0.48	0.18	0.37	0.48
CLIP-L	0.35	0.52	0.32	0.41	0.54
SigLIP-B	0.22	0.49	0.22	0.37	0.52
DeiT-B (IN-1k)	0.22	0.56	0.23	0.38	0.54
ViT-B (augreg, IN-1k)	0.30	0.44	0.40	0.42	0.48

Table 11: (continued)

(c) DBpedia Classes (219 leaf classes, 3 levels), a text hierarchy independent of WordNet; gains in pp.

model	supremum, Haar		census of record					
	$\hat{\delta}_{\max}$	excess	$\hat{\delta}_{99.9}$	excess	$r(p)$	trip.	NC H-R	FS H-R
BGE-base	0.122	-0.032	0.082	-0.021	200 (0.005)	0.89	-0.59	+0.05 $\pm$ 0.05
E5-base	0.127	-0.028	0.086	-0.017	200 (0.005)	0.89	-0.43	+0.04 $\pm$ 0.03
GTE-base	0.132	-0.024	0.088	-0.017	200 (0.005)	0.89	-0.82	+0.08 $\pm$ 0.06

Table 11: (continued) (c) Excess under the original supremum reading (3 Gaussian replicates) and under the census of record (Haar null, 99.9th-percentile statistic, 200 replicates; r = replicates above the real value, left-tail p), sibling-triplet agreement, and the metric gains, marginal as the diagnostic predicts. Tree map on the same classes: no configuration degenerates (largest cluster $\leq 10\%$), the embedders recover the true 70-way level-2 partition at ARI 0.21–0.29 in all 6 configurations and agree with each other at cross-model ARI 0.65–0.82; the configuration the criterion selects there (Euclidean, average linkage) is the one that degenerates on ImageNet, which is why the criterion is applied per dataset. (d) Radial ordering of the four levels under the paper’s projection (chance for strict monotonicity = $1/24 \approx 4.2\%$; shuffle controls ≈ 0), sibling triplet agreement (cosine; chance 0.5), and leaf $\hat{\delta}$ with spectrum-null excess.

(d) HierarCaps (1000 four-level caption chains): a multi-level text hierarchy with no class set.

model	$\rho(\text{level, radius})$	% monotone	trip. L1	trip. L2	$\hat{\delta}$ leaves	excess
BGE-base	+0.80 $\pm$ 0.31	49.6	0.88	0.97	0.130	+0.003
E5-base	+0.87 $\pm$ 0.19	53.9	0.90	0.97	0.139	+0.010
GTE-base	+0.47 $\pm$ 0.55	24.0	0.90	0.97	0.135	+0.006
CLIP-B	+0.64 $\pm$ 0.43	27.2	0.86	0.95	0.123	-0.011
CLIP-L	+0.51 $\pm$ 0.49	18.7	0.86	0.96	0.117	-0.010
SigLIP-B	+0.32 $\pm$ 0.55	9.8	0.83	0.91	0.138	+0.014

Table 11: (continued)

B.11 IS LOCAL STRUCTURE SHARED? PERMUTATION-CALIBRATED AGREEMENT

measure	raw	null mean	$\tau_{0.05}$	calibrated	null-centered	pairs $p < 0.05$
mutual-kNN ($k=10$), Euclidean	0.472	0.010	0.015	0.464	0.462	66/66
mutual-kNN, Poincaré	0.425	0.010	0.019	0.414	0.415	66/66
linear CKA, Euclidean	0.633	0.108	0.111	0.577	0.524	66/66
linear CKA, Poincaré	0.640	0.108	0.111	0.586	0.532	66/66

Table 12: **Cross-model agreement survives permutation calibration.** Means over the 66 model pairs of the 12 vision backbones on the 1000 ImageNet class centroids (census cache). Calibration of Gröger et al. (2026) with $K=200$ permutations of the class correspondence and $\alpha=0.05$, scalar (no layer search): $\tau_{0.05}$ is the $\lceil 0.95(K+1) \rceil$ -th order statistic of the observed score and its nulls (eq. 9), $p = (1 + \#\{\text{null} \geq \text{obs}\})/(K+1)$ (eq. 10), calibrated = $\max\{(\text{obs} - \tau_{0.05})/(1 - \tau_{0.05}), 0\}$ (eq. 12); “null-centered” is the earlier raw-minus-null-mean variant, kept for continuity. The mKNN null mean equals the analytic chance level $k/(n-1) = 10/999$.

B.12 WHAT FOLLOWS FOR PRACTICE? THE COROLLARY

One parameter-free projection and two tasks serve the corollary. The parameter-free map $\phi(x) = \tanh(\frac{1}{2}\|s(x - \mu)\|) \cdot \frac{s(x - \mu)}{\|s(x - \mu)\|}$ centers features, scales the 95th-percentile norm to radius $1/\sqrt{2}$, and preserves directions; prototypes in the ball are means of projected points re-mapped through the origin. Tasks: nearest-centroid (NC) and 5-way 5-shot (FS, 1000 paired episodes); metric arms: Euclidean (R), Poincaré (H), the radial map with Euclidean distances (RT), cosine (COS), and the L2-normalized stack. Changing the readout metric on the intact configuration, which is what the rule selects among, is distinct from deforming the configuration itself.

Both readings predict the gain within datasets. Within each hierarchical dataset the Poincaré advantage tracks the record excess: the nearest-centroid correlation is -0.73 on ImageNet with a confidence interval that excludes zero on all four datasets, the few-shot correlation excludes zero on CIFAR-100, CIFAR-10 and DTD but not on ImageNet, and the depth-test z predicts no gain anywhere (Table 13). The raw reading correlates as strongly (-0.84 on ImageNet for nearest-centroid) and survives partialing out dimension, baseline accuracy, family and spectral covariates; without DINOv2 it holds on ImageNet and CIFAR-100 but not on CIFAR-10/DTD. Pooled cross-dataset correlations are family-driven and not used as evidence; on flat-label datasets no metric helps. Gains carry paired confidence intervals and a McNemar test.

The training objective picks the metric. The radial map with Euclidean distances changes nothing, so the gain is the metric itself, and the split follows the objective: cosine suffices for DINOv2, Poincaré $\approx$ cosine for supervised ViTs, and Poincaré beats cosine for CLIP by up to $+1.3\text{pp}$ even after L2 normalization (Table 13). The rule is read off the families it is scored on: it describes them rather than validating a policy.

A diagnostic, not a policy. Khrulkov et al. (2020) estimate $c \approx 0.33$ from the raw reading of their datasets; the same rule on structureless Gaussian clouds gives the curvatures of Section 6. Cosine is a safe default, gating on the reading is the weakest policy tested, and gains are modest and limited to prototype-based tasks (sample-level tasks never benefit, Table 7). On DBpedia, a three-level text hierarchy, the diagnostic predicts correctly on both sides: genuine excess, no metric gain.

(a) Task accuracies and zero-cost metric gains per cell (10 backbones with cached per-dataset features $\times$ 6 datasets).

model	data	NC acc. (%)		FS acc. (%)		FS advantage over Euclidean (pp)			McNemar
		Eucl.	Poinc.	Eucl.	Poinc.	cosine	radial map	Poinc. _N - cos	NC H vs R
ViT-T	IN	68.1	67.2	97.5	97.6	-0.01	+0.01	+0.08	10^{-22} (-)
	C100	29.3	28.2	62.4	63.1	+1.01	+0.45	+0.55	10^{-5} (-)
	C10	87.5	87.4	87.6	88.1	+0.66	+0.19	+0.29	0.19 (-)
	DTD	63.7	62.5	83.3	83.9	-0.23	+0.32	+0.69	0.03 (-)
	FMNIST	78.5	78.4	81.6	81.9	+0.21	+0.10	+0.31	0.38 (-)
	MNIST	75.9	74.0	74.3	74.6	+0.64	+0.52	+0.20	10^{-13} (-)
ViT-S	IN	77.7	77.5	98.6	98.7	+0.17	-0.02	-0.07	0.09 (-)
	C100	74.9	74.6	92.2	92.4	+0.59	+0.16	-0.00	0.10 (-)
	C10	93.4	93.2	92.0	92.5	+0.56	+0.27	+0.19	0.02 (-)
	DTD	69.3	68.5	85.4	86.0	-0.03	+0.25	+0.73	0.10 (-)
	FMNIST	80.7	80.4	83.7	83.5	+0.28	+0.08	-0.19	0.08 (-)
	MNIST	84.8	84.1	80.5	81.4	+0.34	+0.65	+0.84	10^{-4} (-)
ViT-B	IN	81.6	81.7	98.9	99.0	+0.20	-0.04	-0.05	0.21 (+)
	C100	77.3	77.0	92.5	92.7	+0.56	+0.12	-0.05	0.07 (-)
	C10	94.5	94.0	92.7	93.0	+0.49	+0.18	+0.15	10^{-5} (-)
	DTD	70.8	70.8	85.4	86.2	+0.28	+0.19	+0.69	1.00 (-)
	FMNIST	81.3	81.6	83.4	83.3	+0.03	+0.09	-0.04	0.08 (+)
	MNIST	81.6	80.5	76.8	77.6	+0.19	+0.51	+0.66	10^{-7} (-)
ViT-L	IN	86.1	86.2	99.4	99.4	+0.11	-0.03	-0.02	0.58 (+)
	C100	85.3	85.2	94.8	95.5	+1.17	+0.31	+0.12	0.59 (-)
	C10	97.5	97.5	95.0	95.8	+1.25	+0.35	+0.59	0.76 (+)
	DTD	71.8	71.5	85.6	86.5	+0.13	+0.16	+1.02	0.69 (-)
	FMNIST	83.7	83.5	84.4	84.7	+0.00	+0.17	+0.23	0.12 (-)
	MNIST	84.2	83.6	80.8	81.6	+0.08	+0.42	+0.67	0.00 (-)
DINOv2-S	IN	74.7	74.4	97.4	97.9	+0.68	+0.03	-0.22	0.00 (-)
	C100	73.4	74.4	94.3	95.1	+0.80	+0.15	+0.06	10^{-6} (+)
	C10	95.0	95.2	92.9	93.5	-0.36	+0.13	+1.06	0.14 (+)
	DTD	73.4	73.4	89.8	90.5	+0.30	+0.04	+0.36	1.00 (+)
	FMNIST	82.2	82.2	85.4	85.5	+0.02	+0.07	+0.07	1.00 (-)
	MNIST	83.4	81.9	80.2	80.8	-0.07	+0.45	+0.67	10^{-11} (-)
DINOv2-B	IN	78.9	78.9	97.0	97.8	+1.40	+0.04	-0.63	0.57 (+)
	C100	87.4	87.9	96.5	97.2	+1.11	+0.09	-0.34	10^{-4} (+)
	C10	97.1	97.4	94.7	95.4	+0.27	+0.03	+0.43	10^{-4} (+)
	DTD	74.2	74.7	89.2	90.3	+1.48	+0.01	-0.35	0.31 (+)
	FMNIST	83.2	82.8	85.9	85.9	+0.21	+0.04	-0.18	0.01 (-)
	MNIST	80.6	78.8	76.6	77.1	+0.17	+0.48	+0.47	10^{-16} (-)
DINOv2-L	IN	79.8	80.4	96.2	97.1	+2.13	+0.02	-1.24	10^{-14} (+)
	C100	85.6	87.7	96.1	97.1	+1.80	+0.09	-0.74	10^{-38} (+)
	C10	97.9	98.3	94.1	95.1	+0.92	-0.00	+0.11	10^{-8} (+)
	DTD	74.4	75.2	87.1	88.4	+2.40	+0.03	-0.98	0.11 (+)
	FMNIST	84.4	84.5	87.8	87.9	+0.29	+0.03	-0.03	0.40 (+)
	MNIST	74.1	72.7	70.5	71.1	+0.25	+0.43	+0.46	10^{-12} (-)
DINOv2-G	IN	78.4	78.3	96.2	97.1	+2.57	+0.03	-1.65	0.10 (-)
	C100	89.8	91.0	95.9	97.2	+2.44	+0.13	-1.10	10^{-16} (+)
	C10	97.8	98.2	92.3	93.8	+2.51	-0.02	-1.01	10^{-13} (+)
	DTD	72.9	73.1	87.2	88.5	+2.63	+0.01	-1.38	0.81 (+)
	FMNIST	85.9	85.8	89.0	89.2	+0.38	+0.04	-0.07	0.72 (-)
	MNIST	79.3	78.6	72.9	73.8	+0.30	+0.56	+0.72	0.00 (-)
CLIP-B	IN	61.4	59.9	96.8	96.8	-0.04	-0.00	+0.14	10^{-40} (-)
	C100	65.6	65.1	87.7	88.9	-0.03	+0.44	+1.21	0.07 (-)
	C10	87.9	88.3	89.5	90.1	-0.19	+0.20	+0.90	10^{-4} (+)
	DTD	63.7	63.2	82.2	83.1	-0.23	+0.36	+1.29	0.46 (-)
	FMNIST	75.9	75.1	78.8	78.9	+0.06	+0.17	+0.10	10^{-4} (-)
	MNIST	70.1	66.6	70.9	69.7	+0.04	+0.23	-1.17	10^{-45} (-)
CLIP-L	IN	75.0	74.5	98.4	98.6	+0.01	+0.00	+0.22	10^{-8} (-)
	C100	74.6	74.5	90.0	90.8	+0.12	+0.39	+1.23	0.66 (-)
	C10	95.9	95.9	92.4	93.0	+0.16	+0.25	+0.91	0.59 (-)
	DTD	68.6	69.0	85.2	86.2	-0.08	+0.36	+1.24	0.45 (+)
	FMNIST	76.2	75.9	76.8	77.3	+0.12	+0.33	+0.68	0.09 (-)
	MNIST	85.2	86.1	83.7	84.7	+0.08	+0.58	+1.06	10^{-7} (+)

Table 13: **Both readings predict the zero-cost gain within datasets, the depth verdict predicts nothing, and cosine is the safe default.** (a) Nearest-centroid (NC) and 5-way 5-shot (FS, 1000 paired episodes) accuracy under Euclidean and Poincaré readouts, regenerated from the audited re-runs; FS advantage of cosine and of the radial map with Euclidean distances (the map without the metric changes nothing) over Euclidean, and of the Poincaré readout over cosine after L2 normalization; McNemar p for the NC test-set decisions, Poincaré vs Euclidean (order of magnitude when $p < 10^{-3}$; sign of the NC advantage in parentheses).

(c) Best zero-cost metric per backbone: few-shot advantage over Euclidean, mean over the four hierarchical datasets, and which metric collects it (either: within 0.15pp).

model	best-R (pp)	metric
ViT-T	+0.58	either
ViT-S	+0.49	either
ViT-B	+0.51	either
ViT-L	+0.87	either
DINOv2-S	+0.68	H
DINOv2-B	+1.17	cos
DINOv2-L	+1.84	cos
DINOv2-G	+2.54	cos
CLIP-B	+0.67	H
CLIP-L	+0.65	H

Table 13: (continued)

(d) The same correlations on the raw $\hat{\delta}_{99.9}$, the record excess and the depth-test z (Fisher- z CI95; dm: family-demeaned r).

gain	dataset	raw $\hat{\delta}_{99.9}$ r [CI95]	dm	record excess r [CI95]	dm	depth z r [CI95]
NC H-R	ImageNet	-0.84 [-0.96, -0.45]	-0.83	-0.73 [-0.93, -0.19]	-0.61	-0.28 [-0.77, +0.42]
	CIFAR-100	-0.92 [-0.98, -0.68]	-0.58	-0.79 [-0.95, -0.33]	-0.21	+0.52 [-0.16, +0.87]
	CIFAR-10	-0.63 [-0.90, +0.00]	-0.31	-0.67 [-0.91, -0.06]	-0.34	n/a
	DTD	-0.67 [-0.91, -0.07]	-0.67	-0.76 [-0.94, -0.26]	-0.60	n/a
FS H-R	ImageNet	-0.86 [-0.97, -0.50]	-0.56	-0.21 [-0.74, +0.48]	+0.35	+0.27 [-0.43, +0.77]
	CIFAR-100	-0.62 [-0.90, +0.02]	-0.51	-0.79 [-0.95, -0.31]	-0.80	-0.26 [-0.76, +0.45]
	CIFAR-10	-0.78 [-0.95, -0.31]	-0.64	-0.78 [-0.95, -0.30]	-0.66	n/a
	DTD	-0.87 [-0.97, -0.53]	-0.94	-0.98 [-1.00, -0.91]	-0.97	n/a
FS best-R	ImageNet	-0.83 [-0.96, -0.43]	-0.50	-0.08 [-0.68, +0.58]	+0.47	+0.13 [-0.54, +0.70]
	CIFAR-100	-0.77 [-0.94, -0.27]	-0.75	-0.75 [-0.94, -0.23]	-0.65	+0.22 [-0.48, +0.75]
	CIFAR-10	-0.63 [-0.90, -0.01]	-0.55	-0.62 [-0.90, +0.02]	-0.56	n/a
	DTD	-0.81 [-0.95, -0.36]	-0.78	-0.90 [-0.98, -0.63]	-0.84	n/a

Table 13: (continued) (d) The calibrated reading: the raw 99.9th-percentile statistic, the record excess and the depth-test z of Table 8 as predictors. (e) Validation picking retains the oracle advantage and the zero-validation objective rule modestly beats always-cosine. (f) The gate’s value is avoiding the Poincaré projection on flat-label data; cosine is a safe NC default everywhere. (g) The headline $t = 1/\sqrt{2} \approx 0.71$ was fixed a priori and is conservative for FS.

(e) Few-shot metric-selection policies on held-out episodes (500 validation / 500 test per cell).

policy (test advantage over Euclidean, pp)	all 60 cells	hierarchical (40)
always cosine	+0.23	+0.28
objective rule (zero validation)	+0.28	+0.41
δ -gated objective rule	+0.08	+0.13
validation-picked per cell	+0.63	+0.78
test oracle (reference)	+0.64	+0.78

Table 13: (continued)

(f) Nearest-centroid metric-selection policies.

NC policy (advantage over Euclidean, pp)	all 60	flat (20)	hierarchical (40)
always cosine	+0.48	+0.24	+0.59
always Poincaré	-0.24	-0.70	-0.00
gated rule (flat $\rightarrow$ R)	+0.34	+0.00	+0.51
δ -gated rule	+0.19	+0.00	+0.29

Table 13: (continued)

(g) Sensitivity to the projection target t (36 hierarchical cells).

target t	0.25	0.35	0.41	0.45	0.50	0.58	0.71	0.95
FS adv. mean (pp)	+0.88	+0.87	+0.87	+0.86	+0.84	+0.82	+0.76	+0.56
NC adv. mean (pp)	-0.30	-0.25	-0.23	-0.21	-0.17	-0.12	-0.01	+0.15
% cells FS > 0	97	100	100	100	100	100	100	100

Table 13: (continued)

B.13 PROVENANCE

table	generated file	result file(s)
4	tab_q01_census_final	expR75_census_centered_haar.csv expR52_census_haar_p999_200.csv expR54_census_haar_sup_200.csv expR40b_p999census200.csv expR39c_census200_cache.csv expR57_census_cosine_haar_p999_200.csv expR57_text_cosine_haar_p999_200.csv exp10_local_vs_global.csv expR45_convnet_rows.csv exp20_null_ztable.csv
6	tab_q02_text	expR53_text_haar_p999_200.csv expR53_text_haar_sup_200.csv expR53_text_gauss_p999_200.csv expR48b_text_census200_bs1.csv expR57_text_cosine_haar_p999_200.csv exp18_text_nulls.csv exp4_prompt_variation.csv exp17_wordnet_prompts.csv expR49_template_nulls_bs1.csv expR47_extraction_variance.csv
7	tab_q03_sample_final	expR62_samplelevel_record.csv expR78_khrulkov_replication.csv expR78_khrulkov_replication_summary.csv
8	tab_q04_depth_final	expR56_depth_variants.csv expR69_depth_haarhubs_summary.csv expR64b_wn30bal_summary.csv expR70_inet1k_supervised.csv expR52_census_haar_p999_200.csv expR63_meru_record.csv expR65_hier_finetune.csv expR74_decoupling_summary.csv expR71_meru_radii.csv expR77_positive_control.csv expR77_positive_control_verdict.json expR82_radial_control.csv expR82_radial_control_summary.csv
9	tab_q05_power_final	expR55_depth_power.csv expR55b_depth_power_leafframe.csv expR64b_wn30.csv expR64b_wn30_summary.csv expR64b_wn30bal_summary.csv expR67_frame_choice.json expR69_depth_haarhubs_summary.csv expR80_implanted_alignment.csv expR80_decision.csv expR79_synthetic_deep_poincare.csv expR81_deep_per_backbone.csv expR81_deep_per_backbone_summary.csv expR83_flat_balanced.csv expR83_flat_balanced_summary.csv
10	tab_q06_treemap	exp22_tree_similarity_imagenet.npz exp23_treemap_controls.npz exp23_config_diagnostics.json expR58_treemap_cutfree_summary.csv
11	tab_q07_wordnet_final	exp3_alignment.csv exp28_recovery_per_config.csv exp8_pl_recovery.csv exp1_delta_controls.csv exp8_p6_pooling.csv expR70_inet1k_supervised.csv exp14_dbpedia.csv expR61_dbpedia_record.csv
2	tab_q08_robust_final	exp27_dbpedia_treemap.json exp16_hierarcaps.csv expR72_budget_record.csv expR72_budget_record_summary.csv expR59_imagenet_bootstrap_summary.csv expR73_transfer_bootstrap_record_summary.csv expR66_joint_sensitivity_summary.csv expR60_c_sweep_record.csv analysis4_finetuning.csv
13	tab_q09_corollary	el_delta_by_layer.csv expR66c_joint_sensitivity_summary.csv table1_regenerated.csv exp2_metric_controls.csv exp2b_normalized_stack.csv exp13_mcnemar.csv night/correlation_cis.csv exp20_null_ztable.csv expR68_corollary_excess.csv exp24_val_metric_selection.csv night/t_sweep.csv
3	tab_q10_calibration	exp1_delta_controls.csv expR42_star_calibration.csv
12	tab_q13_local	exp21b_local_global_K200.csv
5	tab_q14_panel	(model panel: stated in the generator; parameter counts from the model cards)

Table 14: Provenance index: every appendix table is generated by script from the released per-cell result files listed here; no table value is hand-transcribed. The same mapping for main-text tables and figures is annotated in the source and shipped with the code release.